

Detailed Project Report (DPR)
On
1.2 MW, Grid connected Power
Generation Project
(Biomass Gasification & Producer
Gas Engine)

At
Plot No 1 Chanaur Industrial Area, Village Chanaur, Tehsil
Dadasiba, District Kangra, Himachal Pradesh – 177113

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EXECUTIVE SUMMARY

Anagram Systems was established in 1994 and is a well-known Furniture manufacturer and specializes and excels in the design, production of customized, chic, trendsetting furniture for the hospitality industry and other living spaces. Anagram's main focus is to work with various Government bodies and mainly Central and State governments for manufacturing and producing bamboo products such as Bamboo Furniture, Bamboo Flooring, Bamboo Doors, and host of other products made from Bamboo. Anagram is working closely with Bamboo design experts to produce products that meet the exact technical and quality standards of the market. Anagram Systems is providing high quality bamboo products, which are ethically sourced, sustainably harvested and designed with utmost attention to detail, we can make a small impact that will ultimately lead to a world of difference.

It is committed to environment protection, customer satisfaction and uphold these core values:

- Produce & supply the best quality bamboo products to our customers & the end users.
- Maintain employee safety and satisfaction.
- Care about our environment and the local communities.
- Learn, Educate & Innovate.

Himachal Pradesh is a hilly state of northern part of India with 55673 km² total geographical area 26.37% is covered with forests having grown bamboo plantation is about 3% of forest area. Eight species of bamboo are found under five genera. The Himachal Pradesh government is promoting bamboo cultivation in arid and semi-arid areas of Kangra, Hamirpur, Mandi and other districts under the National Bamboo Mission to ensure sustainable income to farmers. The agriculture department is promoting Bamboo cultivation among farmers of four development blocks in Kangra district. In 2020-21 the state government under Prime Minister National Bamboo Mission Scheme is planning to plant 90,000 saplings in a 90-hectare area under the setting up of bamboo-related industries under the scheme.

The state of Himachal Pradesh has an annual power generation of 10051 MJ (FY 2021-22) and against the demand of power is 10129 MJ. HP state power installed capacity (MW) about 4007 MW (March 2020) out of which 70% shares is large hydro, the share of RES is about 951 MW combination of solar and small hydro projects.

Anagram System is setting up a State of Art cross laminated bamboo factory producing bamboo board and flooring. The unit on full scale production will utilise 50 MT of raw bamboo per day and during the production process about 30-35 MT per day bamboo waste in form of bamboo dust, chips and nodes will be generated.

Anagram Systems is planning to install a 1.2 MW Gasifier system in combination with producer gas engine at Plot No 1 Chanaur Industrial Area, Village Chanaur, Tehsil Dadasiba, District Kangra, Himachal Pradesh - 177113. As we know Anagram is a bamboo-based furniture supplier and their daily bamboo waste is approx. 30 MT per day. The biomass gasification technology is comparatively a new technology and suitable only for small power plants up to 5MWe, but this technology has a unique advantage of using a variety of biomass. The user can shift from one raw material to another quickly without any change in the plant. Also, gasification technology is a better technology to use municipal solid waste (MSW) briquettes which may be available in plenty in future.

This project contributes to the Sustainable Development through the effective utilisation of surplus biomass (Bamboo) residues available for Power generation, thereby enhancing additional income through rural employment opportunities in the region. Climate Change mitigation through the generation of eco-friendly power and reduce the dependence on fossil fuel based conventional power. This will help to increase the shares of RE sources.

The Ministry of New and Renewable Energy (MNRE), GOI under the waste to energy program on energy from industrial waste streams is promoting setting up Biomass Gasifier for feeding power into the grid or meeting captive power in Industries by giving Central Financial Assistance (CFA). CFA of Rs. 15 Lakhs per 100 kW on prorated basis, or in multiple thereof, will be provided on the gas engine and biomass gasifier systems.

Anagram systems has decided to go for Chanderpur Works Pvt Ltd. for Biomass Gasifiers as they are the most Experienced in India. Chanderpur (CPW) company is a distinguished "Star Export House", by government of India, a National Award Winner for Excellence in Engineering, for the year 2010. Chanderpur works is an approved manufacturer of gasifiers from the Ministry of Non-Conventional Energy Sources, Govt. of India.

PROJECT AT GLANCE

Name	Anagram Systems
Power Plant Installed Capacity	1.2 MW
Location	Plot No 1 Chanaur Industrial Area, Village Chanaur, Tehsil Dadasiba, District Kangra, Himachal Pradesh - 177113
Contact Person	Mr. Amba Charan, Director
Promoter	Mr. Amba Charan, Director
Project Cost	Rs 1337 lakhs
Debt Equity Ratio	70:30
Promoters Equity	Rs 347.1 lakhs (30%)
Term Loan	Rs 809.9 lakhs (70%)
*Subsidy from MNRE	Rs 180 lakhs
Raw Material	Bamboo waste Biomass Pellets
Cost of Raw Material	Rs 3000/ton
Annual escalation on raw material cost	5%
Quantity in Tons per year	10098
Gasifier Model	CPW - 2X750 L/HE
Engine Make	CNPC JICAI POWER COMPANY LIMITED
Rated Engine capacity	300 KW X 4 no.
Water Requirement	96179 litre/hr
Total number of hours	8766
Plant Load Factor (PLF)	85%

Annual Power Generated (inKWh)	8415000
Annual Power available for export during season (KWh)	7573500
Annual Saleable Power (in lakh Rs.)	Rs 588.46 Lakhs during 1 st year of operation
Interest on term loan	9%
Interest on Working Capital	12.5%
Cost of Generation during 1st year of operation	Rs 6.36
Selling Price to HPSEBL	Rs 7.77
Annual escalation on selling price	3.5%
<i>Financial Ratios</i>	
Average DSCR	1.39
BEP (% to Sales)	68%
Cash BEP (% to Sales)	38%
ROE	16.96%
Repayment Period	15 Years

(*subject to approval of scheme)

The total cost of the project is estimated to be Rs 1337 lakhs. Out of this, it is estimated that Rs. 180 Lakhs will be capital subsidy (subject to approval of the scheme for financial year 2021-22). The balance project cost an amount of Rs. 347.1 lakhs 30% will be the promoter's contribution (Equity) and the remaining amount will take the loan from the financial institutions. For financial analysis banks interest is around 9% with a repayment period of 15 years.

The plant is projected to operate for 330 days annually with 85% PLF from first year itself and onwards. The average consumption of Bamboo waste (Pellets) is around 1.2 kg per unit of electricity output and bamboo waste requirements is 10098 MT annually. The price of the Bamboo waste (Pellets) is considered to be Rs. 3000 per ton, considering bamboo waste is converted into pellets at project site. For financial analysis, it is expected that the price of Bamboo waste pellets will increase by 5 % every year. The auxiliary power consumption is 10.0 % of the total generation and line loss considered to be 2% of the units transmitted. It is estimated that the plant will be able to sell the power at Rs.7.77 per unit with annual escalation of 3.5% is assumed for which the promoter will sign a PPA agreement with the local utility on a long-term basis.

This project is very cost sensitive. Sensitivity analysis is carried out with three different cases

- (1) The increase in raw material cost by 10%
- (2) Decrease in revenue by 10%
- (3) The combined effect of Increase in raw material cost by 10% and decrease in revenue by 10%.

The results of these analyses for all three cases are financially viable for this project.

Overall, the project is well conceived and conceptualized, with techno-commercial and financial viability. The expected financial returns are quite satisfactory. The project is being implemented by promoters having requisite background and experience and with proposed employment of experienced professionals, experts, and consultants. All perceived risks have adequate safeguards.

The successful implementation of the project would imply the future development of several biomass-based projects in the state which would benefit all stakeholders.

INTRODUCTION

BACKGROUND

1.1 Description of the Business

The global bamboo market in 2019 was estimated to be more than 60 billion USD (per INBAR) and is likely to reach close to 100 billion USD in the foreseeable future. Bamboo, being the fastest growing plant in the world, promises to be one of the best renewable resources that could potentially substitute hardwood and softwood species. Besides the many traditional applications, in which bamboo in its natural form (stem) is used, a wealth of new bamboo products has become available through industrial processing.

Anagram systems see a tremendous opportunity in “ply bamboo”, which is a natural wood substitute and is classified as “Non-Wood Forest Product” (NWFP) due to its inherent properties. Ply bamboo is gaining popularity in mid and high-end projects and renovations all over the world. Ply bamboo is an eco-friendly, sustainable alternative to traditional plywood and hardwood products. Bleached and carbonised bamboo looks aesthetically pleasing and therefore does not need an overlay of laminate or veneer.

At Anagram systems, plan to set up a world-class integrated plant in Himachal Pradesh (HP) that will be a state-of-the-art manufacturing facility, to produce various sizes of cross laminated bamboo (CLB) made of multiple layers of bamboo strips. The strips are kiln dried, sanded smooth and then laminated edge to edge to create a single-ply panel. These panels are then laminated at right angles to each other to create a multi ply cross laminated bamboo plywood.

The multiple layers are heat pressed together with glue and cured. In line with our commitment to produce environmentally friendly plywood, we are planning to use Polyurethane (PUR) glue to bond the different layers in the plywood. Anagram intends to target ISO 14001 certification which specifies the requirements for an environmental management system that an organisation can use to enhance its environmental performance in a systematic manner that contributes to the environmental pillar of sustainability. Anagram is fully committed to provide aesthetic, durable, eco-friendly, and high-quality bamboo products at optimum price to our customers.

The management team will ensure that Quality Management Systems (QMS) objectives, including those needed to meet requirements of our products are established at relevant functions and levels within the company. The Management team will review the company's Quality Management Systems, at planned intervals, to ensure its continuing suitability, adequacy, and effectiveness. All the processes related

to Quality Management Systems of the company are reviewed by the Management Review Committee. The review includes assessing opportunities for improvement and the need for changes to the Quality Management Systems, including the Policy and objectives. Records of management reviews, assessments and audits will be maintained as part of company records. Safety will be a core value at Anagram and efforts will be made to ensure systems and processes are in place to ensure utmost safety of all employees on site, any contractors and any visiting guests, suppliers, or clients. Safety in design will be a key consideration in the design of the building and our facilities. All staff will be trained at the start of employment in safety policies and practises and on a regular basis throughout their employment with us.

The company plan to take a phased approach from building a facility to the full-scale integrated ply bamboo / biomass electricity plant. The aim is to work diligently in each phase to develop and optimise the efficiency of the facility, while minimising wastage.

Table 01: Phase approach of Bamboo processing unit

Phase-1	Cross laminated bamboo (CLB) factory producing bamboo boards, and predominantly bamboo flooring -We will also undertake model bamboo plantations of different species to be showcased to the local farmers to encourage them to undertake highly profitable High density, industrially processable bamboo plantations
Phase-2	Full scale Ply bamboo Factory producing bamboo ply and clefts for cricket bats also Biomass Charcoal/ Briquetting plant to utilise the bamboo waste generated in plant
Phase-3	Full-scale CLB factory producing Bamboo Boards and the phase 1 factory will keep on producing bamboo flooring only -Train and skill local workers to produce alternative bamboo products using bamboo unusable in board manufacturing -Skill development centre will be made operational
Phase 4	Full scale CLB factory for Clefts of Cricket Bats
Phase-5	Full scale bamboo furniture factory to produce wardrobes, kitchens, beds etc. Biomass Gasification & Electricity Generation plant -Explore other ways to utilise bamboo fibre (pulp & paper, bamboo plastic composite, fibre boards etc.

COMPANY PROFILE

2.1 PROMOTERS

Anagram Systems is promoted by Mr. Amba Charan is a well-qualified and experienced industrialist. Mr Charan has over 30 years of experience in the furniture industry. He is instrumental in bringing modern CNC technology to India in wood and panel processing. He is founder of Anagram Systems, which is a pioneer in modular systems furniture for offices and home segments. He is responsible for new product design, setting up production processes and marketing functions. Prior to Anagram Systems, he was sales executive in the Godrej Delhi branch.

2.2 NAME, ADDRESS, PHONE & EMAIL OF COMPANY

Table 02: Contact details of the company

Name	ANAGRAM SYSTEMS
Address	C-7 ARUNA PARK, LAXMI NAGAR, DELHI
Phone	+91-9717991241, 9205210070
Email	ambacharan@gmail.com

2.3 BUSINESS TYPE

Supplier of Bamboo Based Products (Bamboo boards, Bamboo flooring, Bamboo ply, cricket, and sports equipment) and consulting services to the Indian market.

2.4 KEY MANAGEMENT TEAM & BOARD

The partners of this venture have come together with great passion to raise the awareness and use of eco-friendly bamboo products. They come from a variety of educational and complementary professional backgrounds that bring unique international exposure to success of this project. Key management and operations decisions will be taken together by the Board comprising all the partners. In due course, the company will be structured, and specific roles will be taken up by the team.

Mr Amba Charan – Director & Production Manager

Col Sanjeev Prashar (Retd) – Operations and Farmer Relations

Mrs. Nupur Bhandari – Finance & HR

Mr. Kamlesh Vashisht – Sales and Marketing

Dedicated, skilled, professional, and passionate employees will be a key ingredient of this venture that will help separate from the rest. As projected, Full time and Part time employees listed here are only estimates, this number will vary at different times depending on our implementation schedule and is intended only as a guideline

Indirect Employment: There will be tremendous employment generation at the local level for growing, harvesting, and processing raw Bamboo. We expect that there will be employment of 30,000– 40,000 person days per year from their project.

2.5 NAME & DESIGNATION OF AUTHORISED SIGNATORY

Table 03: Name and Designation of authorised signatory

NAME OF AUTHORISED SIGNATORY	Mr. Amba Charan
DESIGNATION	Director & Production Manager

THE STATE OF HIMACHAL PRADESH

3.1 HIMACHAL PRADESH

Himachal Pradesh is in the northern region of India. The state is surrounded by Jammu and Kashmir on the north, Punjab on the west and the south-west, Haryana on the south, Uttar Pradesh on the south-east and China on the east. **At current prices, Himachal Pradesh's gross state domestic product (GSDP) is estimated at Rs. 1,72,174 crore (US\$ 23.58 billion) in FY22, an increase of 3% YoY.** The state's GDP is estimated to increase at a **CAGR of 7.06%** between 2015-16 and 2021-22. Tertiary sector witnessed the **fastest growth at a CAGR of 10.76% between 2011-12 and 2020-21.**



Fig. 1 State Map

Total exports from Himachal Pradesh stood at US\$ 1,392.91 million in 2019-20. In FY21, exports from Himachal Pradesh stood at US\$ 1,661.89 million.

As of April 2021, Himachal Pradesh had a total installed power generation capacity of 4,155.13 MW, comprising 1,626.39 MW under private utilities, 1,062.21 MW (state utilities) and 1,466.53 MW (central utilities). Himachal Pradesh has immense potential for hydropower as it is naturally blessed with abundant streams and rivers flowing down from towering mountains.

In November 2020, the Union Cabinet approved a proposal for an expenditure of Rs. 1,810 crore (US\$ 245.41 million) for the 210 MW Luhri Stage-I hydropower projects in Himachal Pradesh on the Satluj River. 758.20 million units of electricity annually will be generated by this project. The project will help create ~2,000 direct and indirect jobs and will contribute to the state's overall socioeconomic growth.

The State Government of Himachal Pradesh offers a wide range of policy and fiscal incentives for businesses under the Industrial Policy, 2017. The state has also undertaken other policies such as Tourism Policy 2005 and IT Policy 2001 for upgradation of these sectors in the state. Himachal Pradesh has announced a Revised Solar Policy 2016 under which it plans to establish 700 MW of solar photovoltaic cell capacity.

Under State Budget 2020-21, the highest increase in allocations was observed in Water Supply, Sanitation, Housing and Urban Development at 33% of the overall budget.

PROPOSED LOCATION OF THE POWER PLANT

4.1 BASIS OF SITE SELECTION

Anagram systems planned site for the proposed project at Village: **Chanaur** District: **Kangra, Himachal Pradesh**, which is located near the raw material source to save on the transportation cost. Bamboo is a hollow product and takes a lot of volume to be transported, therefore location of the plant near bamboo growing areas is a prerequisite for the success of the project. The harvest will ideally be picked up within 50 km radius from the plant location.

The Map indicating the project site is appended below for reference –

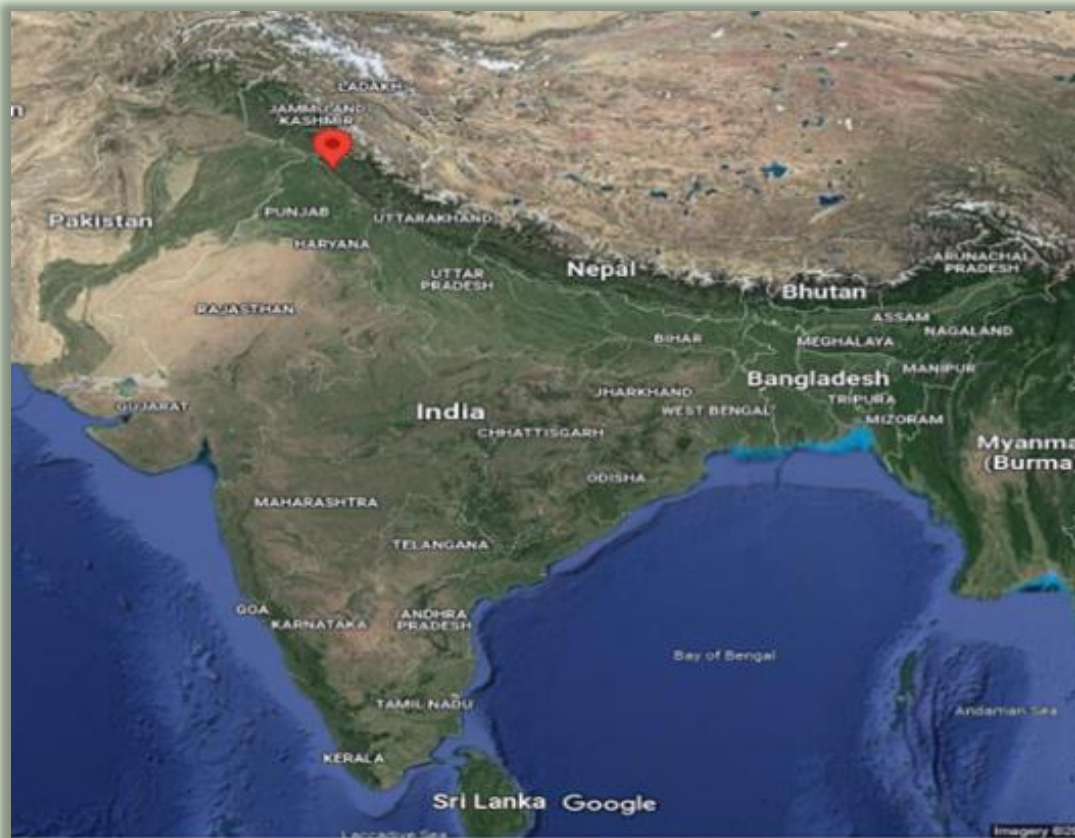


Fig. 2 Project site location

4.2 ABOUT CHANAUR

According to Census 2011 information the location code or village code of Chanaur village is 008715. Chanaur village is in the Indora Tehsil of Dharamshala district in Himachal Pradesh, India. It is situated 104km away from district headquarter Indora. Indora is the sub-district headquarter of Chanaur village. As per 2009 stats, Chanaur is the gram panchayat of Chanaur village.

The total geographical area of the village is 316.78 hectares. Chanaur has a total population of 1,708 people. There are about 343 houses in Chanaur village. As per 2019 stats, Chanaur village comes under Indora assembly & Kangra parliamentary constituency. Pathankot is the nearest town to Chanaur. In Chanaur village, the population of children aged 0-6 is 127 which makes up 10.95 % of total population of the village. Average Sex Ratio of Chanaur village is 1028 which is higher than Himachal Pradesh state average of 972. Child Sex Ratio for the Chanaur as per census is 868, lower than Himachal Pradesh average of 909.

Chanaur village has a higher literacy rate compared to Himachal Pradesh. In 2011, the



Fig. 3 District Map of Kangra

literacy rate of Chanaur village was 89.16 % compared to 82.80 % of Himachal Pradesh. In Chanaur Male literacy stands at 93.85 % while female literacy rate is 84.69 %.

4.3 AVAILABILITY OF WATER

The project requires water for cooling & cleaning of gas, cooling of gas engines, gasifiers. The water will also be required for personal uses of manpower. Most of the equipment will use a water recirculation system, which will reduce the requirement in

the process. So, make-up water will be required. The quantity of water required is 96179 litres/hr. The plant will have a submersible water pump to meet their requirement of water. As only make-up water is reqd. So, there will not be any water effluent.

4.4 AVAILABILITY OF LAND

Anagram has already acquired 54154 m² of land at Village: – Chanaur District Kangra, Himachal Pradesh, for setting up this power project. Gasifiers 2 nos with 4 producer gas engines will be installed in this site only. This land has infrastructural facilities of water, power evacuation systems.



Fig. 4 Satellite view of project site

4.5 AVAILABILITY OF RAW MATERIAL

The plant will be suitably located near the raw material source, which is Kangra District and Hamirpur District, where maggar bamboo is available. Additionally, bamboo is readily available in surrounding districts and can be procured if required. Anagram will actively connect with various farmers and local associations to build a network of local bamboo suppliers. However, Anagram will aim to foster a mutually symbiotic relationship with the local communities by providing support, encouragement, and incentive for the farmers to grow bamboo in huge quantities.

CHOICE OF TECHNOLOGY AND DETAILS

For the first project of this nature, the developer proposes to sell the total generation of power to the state utility HPSEB at the tariff of Rs.7.77/KWh for the 1st Year of operation as per CERC tariff order 2021-22. Details and breakup (fixed + variable) of the tariff is mentioned in the Annexure attached herewith.

5.1 FIRMING OF EQUIPMENT/TECHNOLOGY

Anagram systems has decided to go for Chanderpur Works Pvt Ltd. for Biomass Gasifiers as they are the most Experienced in India. Chanderpur (CPW) company is a recognized "Star Export House", by government of India, a National Award Winner for Excellence in Manufacturing, for the year 2010. CPW is the first in India, who got CE certification for gasifiers, by virtue of incorporating all the safety features as per European Union guidelines. CPW is the first and only company in India, whose gasifiers are tested by third parties, such as Indian Institute of Technology (IIT), New Delhi/ Mumbai, as per government of India's guidelines. Based on which, we have become approved manufacturers by the Government of India and due to which, our customer becomes eligible for getting subsidy from Government of India. Other manufacturers have got approved manufacturers status only by the virtue of the number they have supplied, but their gasifiers are never tested by third party institutes like IIT as per official guidelines.

CPW is the oldest company among all the gasifier manufacturers in India, being established in 1962. CPW is the biggest company in terms of turnover, infrastructure, manpower etc, among all the gasifier manufacturer companies in India and has presence all over the globe. CPW has wide experience of running gasifiers for the last 14 years using different raw materials and in different atmospheric conditions.

Continuous development and improvements in product design have been a corner-stone of the company's policy and these have helped it retain the position of a pioneer and market leader in the field of biomass gasification. All this has been possible because of an able and focused leadership, state-of-the-art facilities (to undertake in-house research and development) and the employees who form the CPW Family.

Several patented designs have been developed and wide recognition achieved in the form of National Award for Technology Development as also several awards for excellence in this field. But CPW understands that technological excellence alone is not enough to succeed in the marketplace.

Biomass gasification is basically conversion of solid fuels (wood/ wood waste, agricultural residues, briquettes. & Pellets etc.) into a combustible gas mixture normally called Producer Gas. The process is typically used for various biomass materials, and it involves partial combustion of such biomass. Partial combustion process occurs when air supply (O_2) is less than adequate for the combustion of biomass to be completed.

The gasifier is essentially a chemical reactor where various complex physical and chemical processes take place. Biomass gets dried, heated, pyrolyzed, partially oxidised and reduced in this reactor as it flows through its four distinct processes take place in a gasifier:

- Drying of the fuel.
- Pyrolysis
- Combustion
- Reduction

Although there is a considerable overlap, each process can occupy a separate zone, in which fundamentally different chemical and thermal reactions take place. The fuel must pass through these zones to be completely converted. The essential characteristics of the down-draft design being used in CPW Gasifiers is that it is designed so that the tars given off in the pyrolysis zone are drawn through the combustion zone, where they will be broken down or burned. When this happens, the energy they contain is usefully recovered and the mixture of gases in the exit stream is relatively clean.

BIOMASS GASIFICATION TECHNOLOGY

Gasification is basically conversion of solid fuels (waste, agricultural residues etc.) into a combustible gas mixture normally called Producer Gas. The process is typically used for various waste materials including Bamboo residues, and it involves partial combustion. Partial combustion process occurs when air supply (O_2) is less than adequate for the complete combustion of biomass/Waste.

6.1 BIOMASS BASED DENSIFICATION TECHNOLOGIES AND FUELS

Densification is processed to convert low bulk density biomass residues into uniform briquettes/pellets form for utilization as energy source. Many biomass residues cannot be used efficiently, as they have low bulk densities. For example, the bulk density of bamboo residues is typical lies in the low range of $50\text{--}200\text{ kg/m}^3$ as compared to high bulk density of wood. This necessitates huge storage space, creates difficulty in handling, energy conversion system and higher transportation cost. All these factors make them economically and efficiently unviable for their widespread utilization. Also, low bulk densities and loose nature of available biomass are associated with faster burning of fuels, resulting in lower operating thermal efficiencies and emissions in the form of fly ash or particulates in the atmosphere. This makes them poor quality biomass fuel.

To improve the quality and exploit this renewable and sustainable source of energy, preprocessing becomes necessary. Densification of biomass by using briquetting or pelletizing technology is the good option to improve the quality of biomass, which compresses loose biomass into briquettes or pellets. This increases the bulk density of biomass and reduces transportation and storage cost and improves the quality of biomass so that it could be used as a better combustible fuel. Thus, in the proposed plant bamboo residues is pre-processed to convert into pellet form and subsequently fed into biomass gasification system for production of electricity.

6.1.1 Biomass Gasification Technology: The Basic Principle

Biomass gasification involves partial combustion of biomass under controlled air supply, leading to generation of producer gas. It is a thermo-chemical (chemical and heat) process in which solid biomass (wood, charcoal, rice husk etc.) is converted into a gaseous fuel by a series of processes including drying, pyrolysis, oxidation, and reduction.

1. Drying is the removal of moisture.
2. Pyrolysis is the thermal decomposition of biomass at high temperatures (greater than 200°C) in the absence of air. The end product of pyrolysis is a mixture of solids (char), liquids (oxygenated oils), and gases (methane, carbon monoxide, and carbon dioxide) with proportions determined by operating temperature, pressure, oxygen content, and other conditions.

If atmospheric air is used as the gasification agent which is the normal practice – the producer gas will consist mainly of carbon monoxide (CO), hydrogen (H₂), methane (CH₄), nitrogen (N₂), and carbon dioxide (CO₂). The first three of these are combustible gases. A typical composition of the gas obtained from wood gasification on volumetric basis is as follows:

Carbon Monoxide	:	18% – 22%
Hydrogen	:	13% – 19%
Methane	:	1% – 5%
Heavier hydrocarbons	:	0.2% – 0.4%
Carbon dioxide	:	9% – 12%
Nitrogen	:	45% – 55%
Water vapour	:	4%

The calorific value (the quantum of energy per given mass) of the producer gas is about 1000 – 1200 kcal (kilocalories)/m³. Producer gas can be used for the generation of motive power either in dual fuel engines (where gas and diesel are mixed and combusted together in the engine) or in diesel engines that have undergone some modification. Engines operating on a spark-ignition system (e.g petrol engines) can be made to run entirely on producer gas, whereas those using compression ignition (CI) systems (e.g., diesel engines) can be made to operate with about 60per cent– 80per cent diesel replacement by the gas. A rough performance ratio for electricity produced from gasifier-based systems is:

- ~ 0.9 – 1.1 kg of biomass/kWh in the dual fuel-mode of operation
- ~ 1.2 – 1.5 kg of biomass/kWh in 100per cent producer gas engine operation.

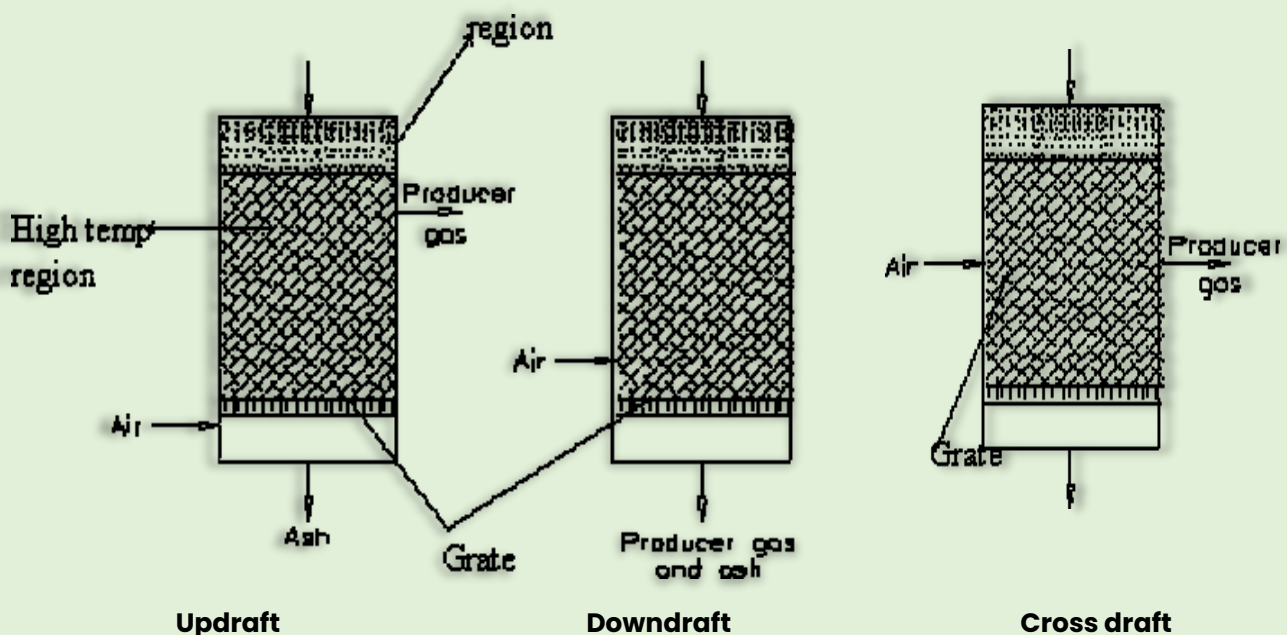


Fig. 5 Types of gasifiers

In all gasifiers, the gasification of solid fuels containing carbon, such as wood, agro-residues etc. takes place in an air-sealed chamber, under a slight vacuum or pressure. The fuel column is ignited at one point and exposed to a continuous air blast during operation, with the producer gas being drawn off at another location. Depending upon the positions of the air inlet and gas withdrawal, three broad types of gasifiers have been designed and operated to date:

- (i) Updraft
- (ii) Downdraft
- (iii) Cross-draft gasifiers.

6.1.2 Biomass Gasification

In a gasifier, solid biomass fuels (wood, agriculture residues, briquettes, etc.) are converted into gaseous fuel (producer gas) by a series of thermo-chemical processes. The producer gas has calorific value of about 1,000 – 1200 kcal/Nm³ and consists mainly of carbon monoxide, hydrogen, and nitrogen. One kilogram of biomass produces about 2.5 Nm³ of producer gas. The producer gas can be used as fuel in IC engines for producing mechanical power and electricity. About 1.2 to 1.5 kg of biomass is required for producing 1 kWh of electricity (using a 100% producer gas engine).

6.2 BIOMASS GASIFICATION TECHNOLOGY IN INDIA

Gasification of coal and biomass has been known for over a century. These technologies were mainly used during the Second World War. However, modern biomass gasification technology has reached maturity only in the last few years. India is one of the few countries having an active research and demonstration programmed on small-scale biomass gasification technologies.

India is now an acknowledged world leader in conversion of solid biomass into clean gaseous fuel. The programmed started in the 1980's has resulted in availability of different gasifier packages having capacities ranging from 5 kW to 2000 kW. The gasifier packages for various applications include those for electricity generation, by running an engine-generator on producer gas; mechanical shaft power when producer gas is used as a fuel in an engine and the engine is directly coupled to the appliance e.g., water pump, grinder, etc.; or thermal applications in which the producer gas is directly burnt in a stove, furnace, or a boiler.

6.3 FIXED BED BIOMASS GASIFIER TECHNOLOGY

The line diagram of the general scheme of a biomass gasifier-based power system explaining the working arrangement is shown in Figure 5. A gasifier-based power generation system consists of a biomass gasifier that converts the solid fuel into

combustible gas. The hot combustible gas coming out the gasifier carries impurities like dust particles and tar vapors, which needs to be cleaned in the cleaning train before supplied to the IC gas engine for power generation. In general, the biomass gasifier is coupled to a series of filters such as gravity filters, wet scrubbers, cyclone separators and bag house filters. A paper filter is used as a safety filter to ensure that the clean gas with permissible levels of tar and dust particulate supplied to the engine. The gas and air mixture are introduced at the air intake manifold of the engine. PID controlled valves are provided both for the air and the gas in order to vary the quantity of gas and air to obtain the optimum mixture.

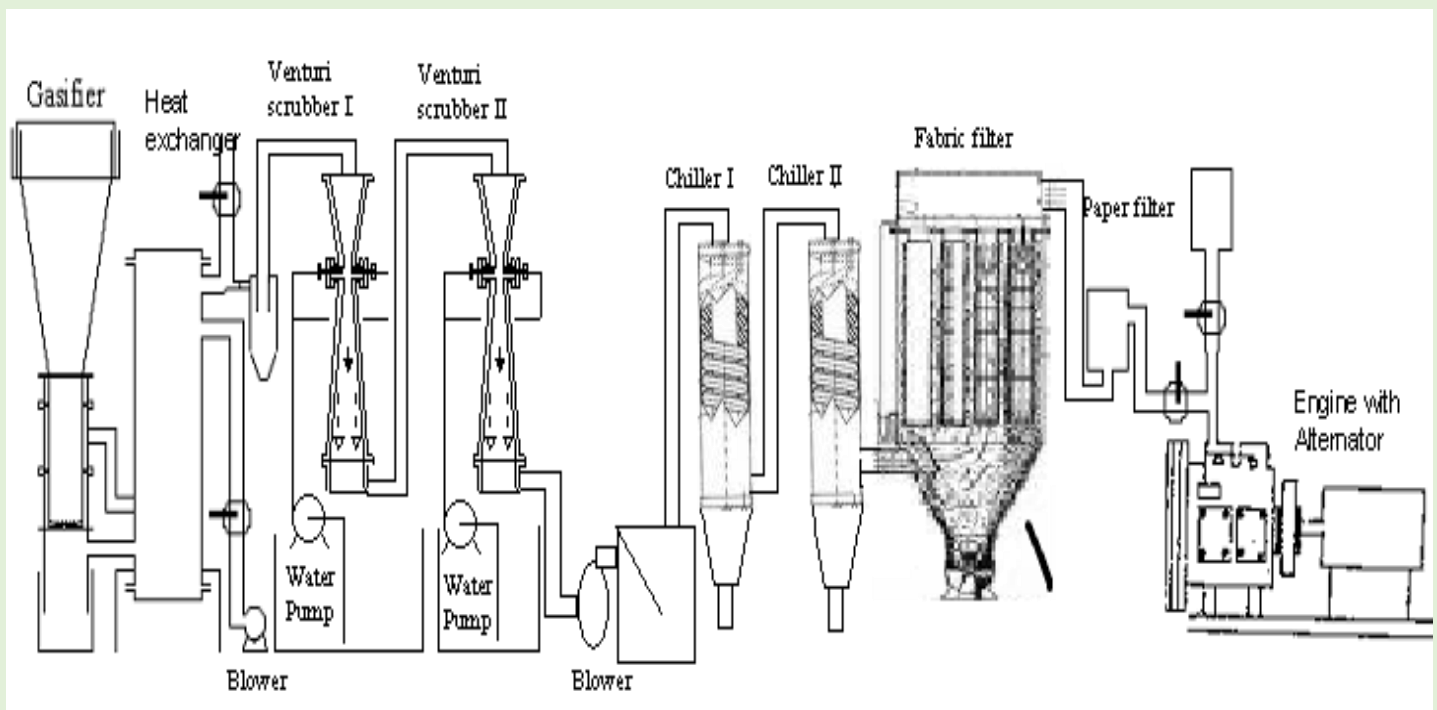


Fig. 6 General scheme of biomass gasifier-based power system

The gas-air mixture is internally supplied to a natural gas engine, which converts the chemical energy in the gas to mechanical energy by rotating a shaft. The engine shaft is in turn coupled to the shaft of an alternator that converts the mechanical energy into electrical energy. The electrical energy so produced by the alternator is distributed through electrical conductors to the connected load to power the bulbs, motors & other electrical appliances or supplied to grid.

6.4 ENERGY CONVERSION

Gasification process converts the renewable fuel into co-products which are themselves fuels. The power derived from gasification and combustion of the resultant products is a source of renewable energy if the gasified compounds were obtained from renewable fuel.

Energy conversion of gasification products is achieved using technological solutions that are best suited to the characteristics of each fuel stream:

- Syngas, the most relevant stream, is converted into power and heat through a high yield cogenerate. (Internal combustion engine)
- Char and pyrolysis oil are converted into thermal energy through a secondary oxidation reactor. (oxidizer)

The typical configuration of a gasification system, in which all streams are converted to produce power and heat, is illustrated below. Gross overall yield of the system in this configuration is around 75–80%. The power produced by converting renewable fuel can be used on site or poured into the national grid to get the incentives granted for the generation of electricity from renewable sources.

6.5 Why Gasification for electricity generation

Combustion releases the energy into high-temperature product gas that is used locally. The product of gasification is, however, a fuel gas. It can be further used for combustion purposes either locally in a furnace or at a distance from the location of gasification in a wide variety of applications like reciprocating or gas engines. The gasification process itself has an efficiency of 75% 90% of conversion to gaseous fuel depending on the fuel, with wood and solid residues leading to efficiencies of 85% to 90% (throughput dependent) and rice husk and high ash fuels around 65% to 70%. The gasification process is based on Otto cycle whereas combustion is based on Rankine cycle.

The specialty of the gasification process is that it can be used for getting electricity even at low power levels (a few kWe to a few MWe) from reciprocating engines at good efficiencies of gas-to electricity (and so biomass-to-electricity), something that would be impossible with combustion team route. To exemplify this feature, at 10 to 200 kWe power level, biomass-steam-electricity would give 3% to 5% efficiency (implying a biomass per kWh consumption of 4.5 to 7.5 kg/kWh) and biomass gasification-reciprocating engine power generation will give conversion efficiencies of 18% to 22% (implying a biomass per kWh consumption of 1 to 1.3 kg/kWh).

The gasification route to energy can be treated as a two-stage combustion process. All the air required normally in a combustion process, typically about six times the biomass throughput, is not to be introduced at once in a gasification process. The gasification process uses about one-third of the air required for combustion and generates cooled combustible gas a little less-than-half of what is generated by combustion process. This gas is then burnt with air about 1.2 to 1.4 times the gas flow rate, with the air flow rate matching with the air required for burning the biomass at stoichiometry.

This approach of staging the conversion process enables the emission abatement process to be dealt with more economically as the flow rates handled are much lower (half of that in the case of combustion). In proposed project gasification process offers distinct advantages due to multi-point control possible without loss of efficiency. So, the proposed plant 1.2 MW on gasification technology is more applicable.

PROJECT FINANCING

7.1 COST OF PLANT AND MACHINERY

The cost of plant & machinery as appraised by management (based on internal evaluation) is INR Rs. 10.90 crores which would include the following

Table 04: DETAILS OF MACHINERY EQUIPEMNTS

S.No.	Item Description
01	Charges for Design, Detail Engineering of Power Plant
02	Charges for Supervision of Installation & Commissioning
03	Skip Charger – This is to feed Biomass from Ground Level to Top of Gasifier Alternatively, this is best done locally
04	State of Art Flare System consisting of • Flare Head with flame arrestor & spark ignition system with auto ignition • Flare Valves assembly with swing valve, pneumatic isolation valve, manual valve & a motorised valve • The middle portion connecting the above two will be as per site requirement or as per local laws
05	Gasifier Cooling Tower – For Clean Water
06	Condensate Neutralisation / Evaporation System – the condensate after neutralisation may still need further treatment to meet the local standards
07	Chiller – As per the requirement
08	100% Producer Gas Engine to give gross peak output of 2000 KWe & continuous duty rating
09	Radiator for Engine Cooling

7.2 CAPITAL COST OF THE PROJECT

The capital cost of the project based on the above cost of plant & machinery, civil works etc and grid connectivity has been worked out as INR Rs. 13.37 Crores as detailed below:

Table 05: Capital cost of the project

S.No.	Particulars	Cost in INR Crores
01	Civil Works including land & land development, approach road, fencing and other ancillary works, local site works like piping, cabling, insulation, sizing/drying equipment, weighbridge, weighing scale, bore well, overhead tanks, vehicles, office set up & other contingencies.	2.00
02	Plant & Machineries including the Gasifier system & its accessories and the Engine Gensets	10.90
03	Misc. Cost like grid connectivity,	0.47
	TOTAL	13.37

7.3 MEANS OF FINANCE

Table 06: Means of finance

Particulars	INR Rs. In Crores
CFA from MNRE	1.8 (13.47%)
Equity /Debt/Bank Financing	11.57 (86.53%)
TOTAL (Project Cost)	13.37 (100%)

7.4 DEBT EQUITY RATIO

The debt equity ratio is considered 70:30, which means 70 percent is debt by bank or financial institutions and 30 percent is equity by the promoter.

7.5 PROMOTERS CONTRIBUTION (EQUITY)

As per the administrative approval of MNRE for the Biomass Gasifier program of year 2020-21, the central financial assistance will be Rs. 15 Lac per 100 KW for Power Plant in Grid Connectivity mode on pro-rata basis or in multiples thereof & thus for this project the CFA works out to be Rs. 3 Crores

Therefore, for the purpose of financing and feasibility calculation, the means of finance for the project has been considered as follows:

Table 07: Promoters Contribution

Particulars	INR Rs. In Crores
CFA from MNRE	1.8
Equity (30%)	3.471
Bank Financing (70%)	8.099
TOTAL	13.37

7.6 TERM LOANS WITH RATE OF INTEREST (FI WISE)

We are in process of finalising the terms & conditions of debt funding with various lenders & a copy of sanction letter will be forwarded to MNRE.

ANY OTHER SOURCE

Company will be utilising its internal accruals for Equity portion and in future fresh Equity is planned to be raised

NAME OF THE LEAD FI (AND OTHER FIs, LENDING TO THE PROJECT)

COMPLETE BANK & BRANCH NAME WITH ADDRESS

BRANCH CODE

RTGS CODE

ACCOUNT NO

TELEPHONE NUMBER

FAX NUMBER

EMAIL

NAME & DESIGNATION OF THE CONTACT PERSON WITH BANK

We are in process of finalising the terms & conditions of debt funding with various lenders & a copy of sanction letter will be forwarded to HAREDA/MNRE after finalisation of the same.

7.7 CONTRIBUTION OF PROJECT ACTIVITY TO SUSTAINABLE DEVELOPMENT

a/ Social well being

- The fuel for this power plant is locally available surplus wood waste. The economy of the local people is improved by selling biomass (wood waste) for the power plant.
- Since the project is in a village it will assist in alleviation of poverty to certain extent by generating both direct and indirect employment in the area.
- Skilled/Unskilled Jobs for regular operation and maintenance of the power plant.

b/ Economic well being

- The biomass gasification process is an alternative to fossil fuel-based power plants and the decentralised power generation through biomass gasification will reduce the transmission and distribution losses.
- The project shall create new rural income resulting from the sales of biomass fuel. Increased income levels shall contribute to the economic security and empowerment of the most vulnerable sections of the society.

c/ Environmental well being

- The project is using bamboo waste for power generation. There is no GHG emission from this project activity. Combustion of biomass waste in the proposed project does not result in net increase in GHG emissions of CO₂, CH₄ and NO_x.

d/ Technology Well being

- The possibility of using the gasifier for internal combustion engines makes it a potential competitor for decentralised power generation. The advantage of decentralised power generation is reduction in transmission and distribution losses and the prospect of rural electrification – a major concern for India.
- Biomass gasification is a cleaner technology where there is no net Greenhouse Gas (GHG) emission.
- The recent development in the gas cooling and cleaning system provides dry producer gas with the tar and particulate level in the range of ppb levels.
- The combined heat and power generation via biomass gasification techniques connected to gas fired engines or gas turbines can achieve significantly higher electrical efficiencies between 22% and 37% compared to biomass combustion technologies with steam generation and steam turbine (15 % to 18%).

In view of the above, the project participant considers that the project activity profoundly Contribute to the sustainable development for the local region as well as to the nation.

7.8 GOVT POLICIES AND PROCEDURE TO AVAIL CFA:

7.8.1 Salient Features of MNRE Scheme:

Ministry of New and Renewable Energy, Govt of India vide their administrative approval had announced the incentives.

Waste-to-Energy projects based on Industrial waste

1. The condition of successful COD of the Waste to Energy projects based on industrial wastes would, inter-alia, imply operation of the project for 3 consecutive months, including continuous operation for at least 72 hours at minimum of 60% of rated capacity or 60% of the quantum of generation as proposed in the DPR, whichever is applicable.
2. Developers shall maintain record of generation from the date of operation of the plant mentioning date, quantity of raw material used daily and gross generation data in suitable units. Provision of suitable meter /system for measuring generation of electricity should be provided in the Waste to Energy plants.

3. Developers ensure that there should not be any emission/leakage of Producer Gas from the plant in the atmosphere.
4. During inspection, Representatives of SNA will witness plant operation to check whether the plant is operating at 80% of rated capacity (or 60% in case of industrial based projects). Accordingly, the SNA team shall prepare a detailed inspection report along with its report on generation data for 3 consecutive months after COD, including performance data for at least 72 hours.
5. Developer shall submit SNA inspection report, SNA authenticated generation data for a consecutive period of 3 months including 72 hours performance data and other requisite documents directly to this Ministry for processing of release of CFA.
6. At this stage, the Bank/FI, if loan is availed, would submit a certificate to the Ministry that the project is not a “Non-Performing Asset” (NPA) and the developer is not a defaulter.
7. The entire CFA will be released to the developer’s loan account maintained in the lending FIs/banks for the purpose of offsetting the loan amount only after successful COD subject to submission of complete documents as per terms & conditions of ‘In- principle’ approval letter, guidelines, and availability of funds.

TECHNOLOGY DESCRIPTION, SYSTEM

DESIGN SPECIFICATION & CAPACITY OF PROJECT

8.1 NATURE OF THE PROJECT AND PROPOSAL

Gasifier and Gas Treatment plant will have an output of 1875 Nm³/hour with calorific value of more than 1050 kcal/Nm³. The max thermal output with Biomass with GCV 4350 Kcal/kg is expected to be 1.875 million Kcal/h.

CPW, India is proposing to supply two units of Gasifier and Gas Treatment plant with cooling & cleaning system, each of 1875 Nm³/hr capacities for reliability in generation. Synthesis gas will be generated from the Biomass as specified. Model is suitable to generate 625 KWH on 100% gas mode.

Below is the technical specification for Chandrapur Works Gasifier Model CPW – 2 x 750L/HE

8.1.1 Specifications for CPW 750 L/HE Gasifier

Model	CPW 750 L/HE
Rated capacity	1875000 Kcal/hr
Gasifier type	Down Draft
Rated gas flow	1875 Nm ³ /hr.
Avg. gas calorific value	1050 Kcal/Nm ³
Ash storage and removal	Wet ash continuous discharge
Startup and operation	Through electrical blower and ignition torch
Fuel type and size	BAMBOO (Bulk Density >350kg/m ³ GCV >3500 kcal/kg) of size 8 to 20 mm
Permissible moisture content	15% maximum
Biomass charging	Online charging
Typical conversion efficiency	>75 %
Maximum Biomass consumption	750 Kg /hr
Typical gas composition (By volume)	CO 18±2%, H ₂ 16±2%, CO ₂ 10±3% CH ₄ up to 3%, N ₂ Balance
Power requirement for gasifier	60 KW connected load
Blowers type	Centrifugal type
Grate shaking mechanism	Rotating grate with 3 HP motor

8.2 PERFORMANCE GUARANTEES

Performance guarantee figures for each Gasifier and Gas Treatment unit as described are presented below

- i. SynGas Volumetric flow rate ($>1875 \text{ Nm}^3/\text{hr}$) at 750 Kg/hr of Biomass consumption
- ii. Calorific value of SynGas ($>1050 \text{ Kcal/Nm}^3$) at 3500 Kcal/kg of Biomass GCV
- iii. Thermal Efficiency ($>75 \%$) for the Gasifier
- iv. The variation in above performance can be $\pm 10\%$

8.3 PROCESS DESCRIPTION OF GASIFIER IN BRIEF

After receiving the sized Biomass from payloader to Hoppers, the belt conveyors feed the material to the bucket elevator and is charged in the gasifier bunker continuously or as and when the bed level goes low. The weighed Fuel is loaded to the bucket elevator and is charged in the gasifier every 12 minutes or as and when the Fuel level goes low. Gasifier is provided with **required amount of air** either through suction in suction mode or by primary fan in pressure mode depending upon the application. The gas passes through a hot air cyclone and then two venturi scrubbers where the gas is scrubbed and, in the process, also cooled to around **40 °C**. It consists of a spray nozzle mechanism which washes the producer gas with a wide sprayed water jet. Water is recycled from primary venturi pit to spray nozzle continuously. After the primary venturi scrubber, the gas passes through secondary venturi scrubber and then to secondary blower. The gas then enters the Tar box and secondary cyclone where the moisture is separated from the cooled gas then to buffer tank whose pressure is maintained by PLC as required. Before feeding to engine gas passes through Coarse, Fine filter and fabric filters for separating moisture and micro fine particulates. For ash removal, the grate is rotate automatically by a grate rotating motor through timer in specific time interval as per load and bed level deviation. In wet ash discharge system, the ash is continuously discharged with the help of a mud pump and then goes to the circulating pond passing through a mesh for screening out ash and char from water. Ash and char will collect through screw conveyor for further disposal. A control system installed with a range of instruments measuring level, temperature, pressures, volume of air and gas at different points of the Gasifier. The Control system helps the control room operator to monitor the complete gasification process and take timely remedial and corrective action.

8.4 GRID VOLTAGE AT WHICH INTERCONNECTION IS TO BE DONE

Essentially for Grid interconnecting a power plant to the grid, the voltage and frequency parameters of the generator need to be synchronised to the grid's parameters

Thus, the hardware needed is as below:

A synchronising relay. One relay is needed per engine. This essentially matches the generator parameters to the grid parameters, a voltage transformer, most engines generate electricity at 415 Volts. We need to step this up to 11000 Volts. Thus, a step-up transformer is needed. The electricity can be exported at 33 KVA to the nearest sub-station which is located 2 to 4 km from the proposed site. The ability to isolate each engine from the grid and the transformer from all the items is a must. This then means the need for various circuit breakers. There are thus circuit breakers after each engine and a circuit breaker after the transformer. With this arrangement, any hardware can be isolated from any other hardware. Various interconnecting cables etc. will be needed.

8.5 VARIABLE LOAD RESPONSE CAPABILITY

While planning, gasification based (100% producer gas based) power generation systems, the nature of connected load (type of load, individual rating as well as ON-OFF frequency etc.) are extremely important. This is because of the following facts:

- These systems work with producer gas that is basically manufactured 'ON-LINE'.
- The response of the gasifier in terms of sudden demand on gas production / conditions requiring sudden reduction in gas production etc. must be considered.
- In addition, the gas flow control system response to sudden variations in load also must be considered.
- The fact that many inductive loads could have high starting current needs to be further considered

It is the endeavour of our company to provide State-of-Art systems with very fast response from the gasifiers as well as the control system. For best performance under variable load conditions, the following recommendations should be considered:

The size of the biggest motor that can be run on the system should not be more than 25% of the stand-alone duty rating of the system.

Table 08: Variable load response capability

Sr. No.	Operating Load (% of stand-alone duty rating)	Permissible Load Variation
1	At 50%	15% of rating in terms of inductive load
2	At 70%	10% of rating in terms of inductive load
3	At 80%	5% of rating in terms of inductive load
4	Above 80%	Only very small loads can be slowly added

8.6 BIOMASS STORAGE AND HANDLING SYSTEM

Before embarking on the storage and handling systems of the bamboo residues, the fuel characteristics analysis was carried out. The following table gives the proximate and ultimate analysis of bamboo waste.

Proximate Analysis

Calorific Value	4350 Kcal/kg
Ash content	1.02 (% by Mass)
Volatile matter	75.20 (% by Mass)
Fixed Carbon	12.81 (% by Mass)

Ultimate Analysis

Carbon %	22.5%
Hydrogen %	3.5%
Nitrogen %	0.95%
Oxygen %	18.6%
Ash	1.02%

As we are planning to use bamboo waste as raw material, bamboo residues handling, and storage system are discussed below:

8.7 BAMBOO PREPARATION SYSTEM

Bamboo residues generally have low bulk densities and because of fibrous nature they frequently have negative repose. These two characteristics make them difficult to store and handle. As biomass fuels have a low calorific value per unit volume their fuel storage system is characterised by their large physical volume compared to equivalent fossil fuel systems.

Keeping in mind the above facts, the wood preparation and handling system of the plant is designed to ensure smooth and uniform flow of bamboo to the gasifier.

Bamboo preparation system of the proposed plant consists of bamboo residue storage area to reduce the moisture content to 10-15% and bamboo residues in chips form to a size of 15 to 25 mm will be available.

8.8 BAMBOO RESIDUES STORAGE AREA

The chipped bamboo will be stored in open yards with an arrangement of a mobile crane, which will spread the bamboo for open air drying to reduce the moisture of wood to 12-15%.

The dry bamboo will be stored in a Covered Storage Yard. The bamboo residues will be taken to a storage yard with the help of a Belt Conveyor. From storage yard to gasifier, bamboo residues will be carried away by a skip hoist.

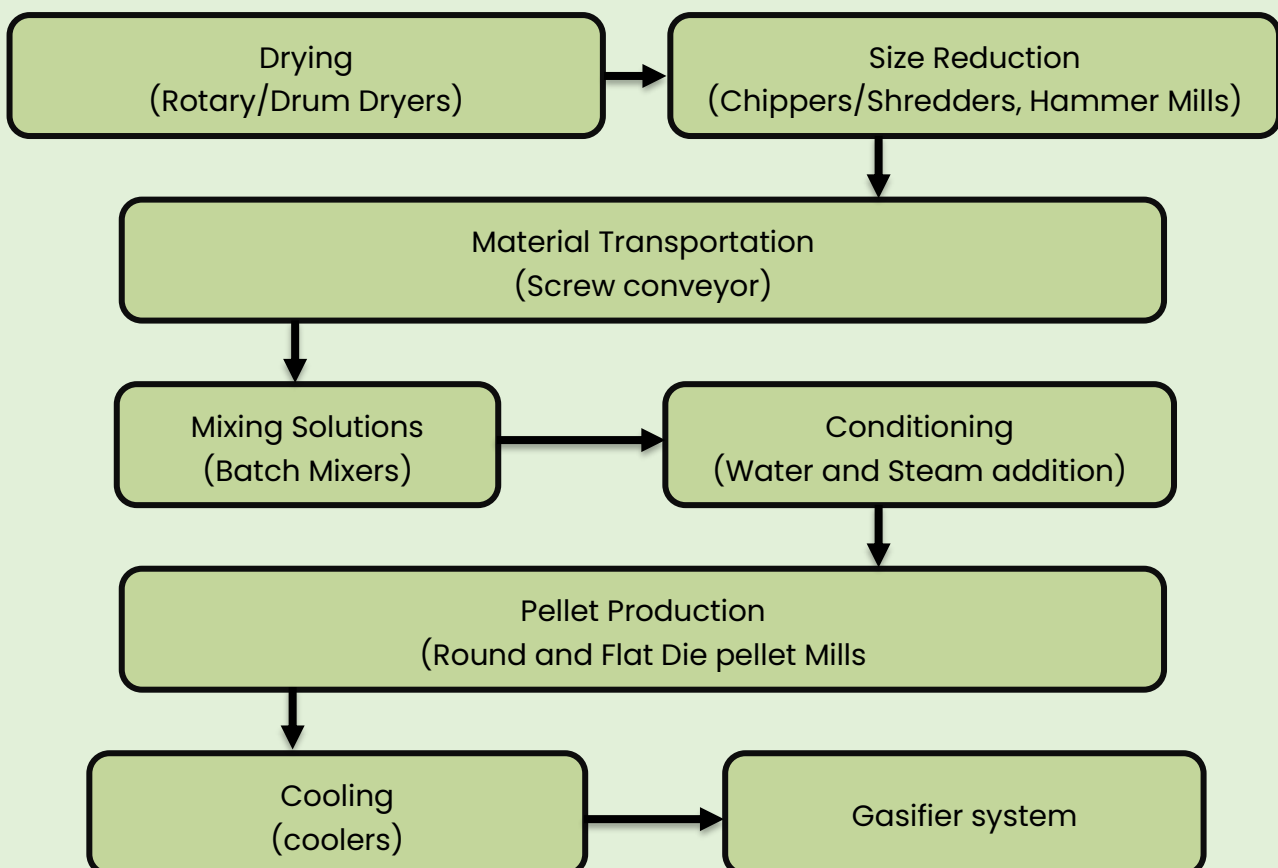
8.9 CHIPPER AREA

The chipper area comprises a raised platform having intermediate log storage space for spread over an area required for the chipped bamboo. The chipped bamboo will be allowed to fall into a covered storage yard. Chipped bamboo from this yard will be transferred to the gasifier with the help of a skip hoist.

8.10 PELLET PRODUCTION

In order to produce pellets, the bamboo residues should be dried using dryers to reduce the moisture content and so that it would be easily processed. The pre-processing would vary depending on the moisture content on it. The dried material undergoes size reduction using hammer mill, chipper, and shredder and the size reduced bamboo material is transported for processing through the conveyor to the mill. Then the conditioned bamboo residues move in the perforated die and roller which rolls on the die compresses the biomass and thus bamboo residues get densified and transforms to pellets. After the process, the pellets are cooled to ambient temperature using coolers or left as such in open space. Once the pellets get cooled, the product undertakes sieving process to remove fines. Then through the bucket elevators, the pellets are fed into gasification plant, or stored in warehouse.

Steps of Pellet Production



CAPITAL COST AND TECHNO-ECONOMIC ANALYSIS

9.1 FINANCING SCHEME

The total cost of the project is Rs.13.37 Crores. It is planned that the promoter will be putting all the project cost after availing the Central Financial assistance from Ministry of Non-Conventional Energy Sources, Govt. of India. The CFA expected is Rs. 180 lakhs, and the balance project cost is Rs. 1157 lakhs.

9.1.1 Promoter's Contribution

- (i) Equity – 30 % of the balance project cost or Rs. 347.1 Lakhs shall be provided by the promoters.
- (ii) Unsecured Loans – 70% of the balance project cost or Rs. 809.9 Lakhs shall be provided by the Bank.

9.1.2 Project Debt

70 % of the balance project cost i.e., Rs. 809.9 Lakhs will be provided by financial institutions.

9.1.3 CFA

Factory is expecting the capital subsidy of Rs. 180 lakhs as Central Financial Assistance.

Table 10: Cost of the Project

COST OF THE PROJECT		
S. No.	PARTICULARS	Amount to be Incurred (Rs Lakhs)
1	Land and Buildings & Civil Works	200
2	Plant and Machinery	1090
3	Power evacuation System (2Km from substation)	15
4	Miscellaneous Fixed Assets	8
5	Interest during Construction Period	8
6	Preliminary and Contingencies	16
	Total Project Cost (Rs.)	1337

Biomass Gasifier Power Projects: Himachal					
S No.	Assumption Head	Sub-Head	Sub-Head (2)	Unit	Parameters
1	Power Generation	Capacity	Installed Power Generation Capacity	MW	1.2
			Auxiliary Consumption during stabilisation	%	10%
			Auxiliary Consumption after stabilisation	%	10%
			PLF (Stabilisation for 6 months)	For First 6 Months	85%
			PLF (during first year after Stabilisation)	Next six months	85%
			PLF (second year onwards)	2nd Year Onwards	85%
			Useful Life	Years	25
2	Project Cost	Capital Cost	Power Plant Cost	Rs Lakh	1337
3	Sources of Fund	Debt: Equity	Debt	%	70%
			Equity	%	30%
			Subsidy from MNRE	Rs Lakh	180
			Total Debt Amount	Rs Lakh	809.9
			Total Equity Amount	Rs Lakh	347.1
		Debt Component	Loan Amount	Rs Lakh	809.9
			Moratorium Period	years	0
			Repayment Period(incld Moratorium)	years	15
			Interest Rate	%	9%
		Equity Component	Equity amount	Rs Lacs	347.1
			Return on Equity (upto 20 years)	% p.a	16.96%
			Return on Equity (after 20 years)	% p.a	21.52%
		Depreciation	Depreciation Rate(Power Plant)	%	4.67%
			Depreciation Rate 16th year onwards	%	2.00%
4	Working Capital	For Fixed Charges	O&M Charges	Months	1
			Maintenance Spare	(% of O&M expenses)	15.00%
			Receivables for Debtors	Months	1.5
5	Fuel Related Assumptions	For Variable Charges	Biomass Stock	Days	10
			Interest On Working Capital	%	12.50%
		Biomass	Specific Fuel Consumption	kg/kWh	1.2
			Base Price	Rs/T	3000
			Biomass Price Escalation Factor	%	5.00%
6	Operation & Maintenance		O&M Cost of power plant	Rs Lakh	65
			Total O & M Expenses Escalation	%	3.84%
7	Generation & Sale of Energy		Total No .Of Hours		8766
		Input data based on CERC Tariff order 2020-21			
		Input data from OEM			
		Input data from Client/IREDA			

Table 11: Input Parameters

Table 12: Schedule of Raw Material

Particulars	Raw Material for Power					
	Bamboo waste as a raw material per unit. (Kg/KWh)	Total no. of units of electricity generated. (KWh)	No. of days	Total Qty. of Biomass required (Kg)	Rate per kg (Including Transportation) (Rs/kg)	Total (Rs. in Lacs)
Year 1 2022-23	1.2	8415000	330	10098000	3.00	302.94
Year 2 2023-24	1.2	8415000	330	10098000	3.15	318.09
Year 3 2024-25	1.2	8415000	330	10098000	3.31	333.99
Year 4 2025-26	1.2	8415000	330	10098000	3.47	350.69
Year 5 2026-27	1.2	8415000	330	10098000	3.65	368.23
Year 6 2027-28	1.2	8415000	330	10098000	3.83	386.64
Year 7 2028-29	1.2	8415000	330	10098000	4.02	405.97
Year 8 2029-30	1.2	8415000	330	10098000	4.22	426.27
Year 9 2030-31	1.2	8415000	330	10098000	4.43	447.58
Year 10 2031-2032	1.2	8415000	330	10098000	4.65	469.96
Year 11 2032-33	1.2	8415000	330	10098000	4.89	493.46
Year 12 2033-34	1.2	8415000	330	10098000	5.13	518.13
Year 13 2034-35	1.2	8415000	330	10098000	5.39	544.04
Year 14 2035-36	1.2	8415000	330	10098000	5.66	571.24
Year 15 2036-37	1.2	8415000	330	10098000	5.94	599.80
Year 16 2037-38	1.2	8415000	330	10098000	6.24	629.79
Year 17 2038-39	1.2	8415000	330	10098000	6.55	661.28
Year 18 2039-40	1.2	8415000	330	10098000	6.88	694.34
Year 19 2040-41	1.2	8415000	330	10098000	7.22	729.06
Year 20 2041-42	1.2	8415000	330	10098000	7.58	765.51
Year 21 2042-43	1.2	8415000	330	10098000	7.96	803.79
Year 22 2043-44	1.2	8415000	330	10098000	8.36	843.98
Year 23 2044-45	1.2	8415000	330	10098000	8.78	886.18
Year 24 2045-46	1.2	8415000	330	10098000	9.21	930.49
Year 25 2046-47	1.2	8415000	330	10098000	9.68	977.01

Table13: Schedule of Sales Realisation

Particular	Raw Material for Power											
POWER GENERATION 1.2 MW												
	Units produ ced for 1.2 MW Hr	No. of Hours	Total no. of units	PLF	No. of days in a year	Total no. of units at full capacit y	Capacit y utilised %	Total no of unit produce d at Capacit y Utilised	Less :- Internal Utilities @ 10%	Units availabl e for export during season	Rate / unit (Based on CERC Tariff Order)	Total (Rs. in Lacs)
Year 1 2022-23	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	7.77	588.46
Year 2 2023-24	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	8.04	609.06
Year 3 2024-25	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	8.32	630.37
Year 4 2025-26	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	8.61	652.44
Year 5 2026-27	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	8.92	675.27
Year 6 2027-28	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	9.23	698.91
Year 7 2028-29	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	9.55	723.37
Year 8 2029-30	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	9.89	748.69
Year 9 2030-31	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	10.23	774.89
Year 10 2031-2032	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	10.59	802.01
Year 11 2032-33	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	10.96	830.08
Year 12 2033-34	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	11.34	859.14
Year 13 2034-35	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	11.74	889.20
Year 14 2035-36	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	12.15	920.33
Year 15 2036-37	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	12.58	952.54
Year 16 2037-38	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	13.02	985.88
Year 17 2038-39	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	13.47	1020.38
Year 18 2039-40	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	13.94	1056.10
Year 19 2040-41	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	14.43	1093.06
Year 20 2041-42	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	14.94	1131.32
Year 21 2042-43	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	15.46	1170.91
Year 22 2043-44	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	16.00	1211.90
Year 23 2044-45	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	16.56	1254.31
Year 24 2045-46	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	17.14	1298.21
Year 25 2046-47	1250	24	30000	85%	330	8415000	100%	8415000	841500	7573500	17.74	1343.65

Table 14: Schedule of Working Capital Requirement

Particulars	Current Assets					Current Liabilities		Working Capital Requirement	Working Capital Funding	Interest on Working Capital @12.5%
	Raw Material stock	Sundry Debtors	O&M Expenses	Maintenance Spares	Sub Total (A)	Liabilities for expenses	Sub Total (B)			
Days	10	45	30			120				
Year 1 2022-23	8.42	73.56	5.42	0.81	88.20	3.45	3.45	84.76	84.76	10.59
Year 2 2023-24	8.84	76.13	5.62	0.84	91.44	3.60	3.60	87.84	87.84	10.98
Year 3 2024-25	9.28	78.80	5.84	0.88	94.79	3.75	3.75	91.04	91.04	11.38
Year 4 2025-26	9.74	81.55	6.06	0.91	98.27	3.92	3.92	94.35	94.35	11.79
Year 5 2026-27	10.23	84.41	6.30	0.94	101.88	4.09	4.09	97.79	97.79	12.22
Year 6 2027-28	10.74	87.36	6.54	0.98	105.62	4.27	4.27	101.35	101.35	12.67
Year 7 2028-29	11.28	90.42	6.79	1.02	109.51	4.47	4.47	105.05	105.05	13.13
Year 8 2029-30	11.84	93.59	7.05	1.06	113.54	4.67	4.67	108.87	108.87	13.61
Year 9 2030-31	12.43	96.86	7.32	1.10	117.71	4.88	4.88	112.84	112.84	14.11
Year 10 2031-2032	13.05	100.25	7.60	1.14	122.05	5.10	5.10	116.95	116.95	14.62
Year 11 2032-33	13.71	103.76	7.90	1.18	126.55	5.33	5.33	121.22	121.22	15.15
Year 12 2033-34	14.39	107.39	8.20	1.23	131.21	5.57	5.57	125.64	125.64	15.70
Year 13 2034-35	15.11	111.15	8.51	1.28	136.05	5.83	5.83	130.22	130.22	16.28
Year 14 2035-36	15.87	115.04	8.84	1.33	141.08	6.10	6.10	134.98	134.98	16.87
Year 15 2036-37	16.66	119.07	9.18	1.38	146.29	6.38	6.38	139.90	139.90	17.49
Year 16 2037-38	17.49	123.23	9.53	1.43	151.69	6.68	6.68	145.01	145.01	18.13
Year 17 2038-39	18.37	127.55	9.90	1.48	157.30	6.99	6.99	150.31	150.31	18.79
Year 18 2039-40	19.29	132.01	10.28	1.54	163.12	7.32	7.32	155.80	155.80	19.48
Year 19 2040-41	20.25	136.63	10.67	1.60	169.16	7.66	7.66	161.50	161.50	20.19
Year 20 2041-42	21.26	141.41	11.08	1.66	175.42	8.02	8.02	167.40	167.40	20.93
Year 21 2042-43	22.33	146.36	11.51	1.73	181.93	8.40	8.40	173.52	173.52	21.69
Year 22 2043-44	23.44	151.49	11.95	1.79	188.67	8.80	8.80	179.87	179.87	22.48
Year 23 2044-45	24.62	156.79	12.41	1.86	195.68	9.22	9.22	186.45	186.45	23.31
Year 24 2045-46	25.85	162.28	12.89	1.93	202.94	9.66	9.66	193.28	193.28	24.16
Year 25 2046-47	27.14	167.96	13.38	2.01	210.48	10.12	10.12	200.37	200.37	25.05

Table 15: Schedule of Depreciation

Schedule of Depreciation				
Particulars	Land & Buildings & Civil Work	Plant & Machinery	Other Fixed Assets	Total
Rate of Depreciation	(As per CERC Tariff Order)		Year 1-15	4.67%
			16th Year onwards	2.00%
Opening Balance	200.00	1090.00	8.00	1298.00
Add :- Interest during construction period	1.34	6.61	0.05	8.00
Total	201.34	1096.72	8.05	1306.00
Year 1				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				58.14
W.D.V.				1196.87
Year 3				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				55.43
W.D.V.				1131.44
Year 4				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				52.84
W.D.V.				1078.60
Year 5				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				50.37
W.D.V.				1028.23
Year 6				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				48.02
W.D.V.				980.21
Year 7				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				45.78
W.D.V.				934.44
Year 8				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				43.64
W.D.V.				890.80
Year 9				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				41.60
W.D.V.				849.20
Year 10				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				39.66
W.D.V.				809.54

Year 11				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				37.81
W.D.V.				771.74
Year 12				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				36.04
W.D.V.				735.70
Year 13				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				34.36
W.D.V.				701.34
Year 14				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				32.75
W.D.V.				668.59
Year 15				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				31.22
W.D.V.				637.36
Year 16				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				12.75
W.D.V.				624.62
Year 17				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				12.49
W.D.V.				612.12
Year 18				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				12.24
W.D.V.				599.88
Year 19				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				12.00
W.D.V.				587.88
Year 20				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				11.76
W.D.V.				576.13
Year 21				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				11.52
W.D.V.				564.60

Year 22				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				11.29
W.D.V.				553.31
Year 23				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				11.07
W.D.V.				542.25
Year 24				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				10.84
W.D.V.				531.40
Year 25				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation				10.63
W.D.V.				520.77

Table 16: Schedule of Interest & Repayment of Loan

Schedule of Interest & Repayment of Loan				
SCHEDULE - I				
Total Term Loan		809.9	Lacs	
Repayment Mode		Monthly		
Repayment Period		15 Years		
No. of Installments		180		
Rate of Interest		9%		(Rs. In lacs)
Particulars	Interest	Principal	Total	Balance of Loan Amount
Year 1				
1	6.07	0.00	6.07	809.90
2	6.07	0.00	6.07	809.90
3	6.07	0.00	6.07	809.90
4	6.07	0.00	6.07	809.90
5	6.07	0.00	6.07	809.90
6	6.07	0.00	6.07	809.90
7	6.07	4.50	10.57	805.40
8	6.04	4.50	10.54	800.90
9	6.01	4.50	10.51	796.40
10	5.97	4.50	10.47	791.90
11	5.94	4.50	10.44	787.40
12	5.91	4.50	10.40	782.90
Total	72.38	27.00	99.38	
Year 2				
1	5.87	4.50	10.37	778.40
2	5.84	4.50	10.34	773.90
3	5.80	4.50	10.30	769.41
4	5.77	4.50	10.27	764.91
5	5.74	4.50	10.24	760.41
6	5.70	4.50	10.20	755.91
7	5.67	4.50	10.17	751.41
8	5.64	4.50	10.13	746.91
9	5.60	4.50	10.10	742.41
10	5.57	4.50	10.07	737.91
11	5.53	4.50	10.03	733.41
12	5.50	4.50	10.00	728.91
Total	68.23	53.99	122.23	
Year 3				
1	5.47	4.50	9.97	724.41
2	5.43	4.50	9.93	719.91
3	5.40	4.50	9.90	715.41
4	5.37	4.50	9.87	710.91
5	5.33	4.50	9.83	706.41
6	5.30	4.50	9.80	701.91
7	5.26	4.50	9.76	697.41
8	5.23	4.50	9.73	692.91
9	5.20	4.50	9.70	688.42
10	5.16	4.50	9.66	683.92
11	5.13	4.50	9.63	679.42

12	5.10	4.50	9.60	674.92
Total	63.37	53.99	117.37	
Year 4				
1	5.06	4.50	9.56	670.42
2	5.03	4.50	9.53	665.92
3	4.99	4.50	9.49	661.42
4	4.96	4.50	9.46	656.92
5	4.93	4.50	9.43	652.42
6	4.89	4.50	9.39	647.92
7	4.86	4.50	9.36	643.42
8	4.83	4.50	9.33	638.92
9	4.79	4.50	9.29	634.42
10	4.76	4.50	9.26	629.92
11	4.72	4.50	9.22	625.42
12	4.69	4.50	9.19	620.92
Total	58.52	53.99	112.51	
Year 5				
1	4.66	4.50	9.16	616.42
2	4.62	4.50	9.12	611.92
3	4.59	4.50	9.09	607.43
4	4.56	4.50	9.06	602.93
5	4.52	4.50	9.02	598.43
6	4.49	4.50	8.99	593.93
7	4.45	4.50	8.95	589.43
8	4.42	4.50	8.92	584.93
9	4.39	4.50	8.89	580.43
10	4.35	4.50	8.85	575.93
11	4.32	4.50	8.82	571.43
12	4.29	4.50	8.79	566.93
Total	53.66	53.99	107.65	
Year 6				
1	4.25	4.50	8.75	562.43
2	4.22	4.50	8.72	557.93
3	4.18	4.50	8.68	553.43
4	4.15	4.50	8.65	548.93
5	4.12	4.50	8.62	544.43
6	4.08	4.50	8.58	539.93
7	4.05	4.50	8.55	535.43
8	4.02	4.50	8.52	530.93
9	3.98	4.50	8.48	526.44
10	3.95	4.50	8.45	521.94
11	3.91	4.50	8.41	517.44
12	3.88	4.50	8.38	512.94
Total	48.80	53.99	102.79	
Year 7				
1	3.85	4.50	8.35	508.44
2	3.81	4.50	8.31	503.94
3	3.78	4.50	8.28	499.44
4	3.75	4.50	8.25	494.94
5	3.71	4.50	8.21	490.44

6	3.68	4.50	8.18	485.94
7	3.64	4.50	8.14	481.44
8	3.61	4.50	8.11	476.94
9	3.58	4.50	8.08	472.44
10	3.54	4.50	8.04	467.94
11	3.51	4.50	8.01	463.44
12	3.48	4.50	7.98	458.94
Total	43.94	53.99	97.93	
Year 8				
1	3.44	4.50	7.94	454.44
2	3.41	4.50	7.91	449.94
3	3.37	4.50	7.87	445.45
4	3.34	4.50	7.84	440.95
5	3.31	4.50	7.81	436.45
6	3.27	4.50	7.77	431.95
7	3.24	4.50	7.74	427.45
8	3.21	4.50	7.71	422.95
9	3.17	4.50	7.67	418.45
10	3.14	4.50	7.64	413.95
11	3.10	4.50	7.60	409.45
12	3.07	4.50	7.57	404.95
Total	39.08	53.99	93.07	
Year 9				
1	3.04	4.50	7.54	400.45
2	3.00	4.50	7.50	395.95
3	2.97	4.50	7.47	391.45
4	2.94	4.50	7.44	386.95
5	2.90	4.50	7.40	382.45
6	2.87	4.50	7.37	377.95
7	2.83	4.50	7.33	373.45
8	2.80	4.50	7.30	368.95
9	2.77	4.50	7.27	364.46
10	2.73	4.50	7.23	359.96
11	2.70	4.50	7.20	355.46
12	2.67	4.50	7.17	350.96
Total	34.22	53.99	88.21	
Year 10				
1	2.63	4.50	7.13	346.46
2	2.60	4.50	7.10	341.96
3	2.56	4.50	7.06	337.46
4	2.53	4.50	7.03	332.96
5	2.50	4.50	7.00	328.46
6	2.46	4.50	6.96	323.96
7	2.43	4.50	6.93	319.46
8	2.40	4.50	6.90	314.96
9	2.36	4.50	6.86	310.46
10	2.33	4.50	6.83	305.96
11	2.29	4.50	6.79	301.46
12	2.26	4.50	6.76	296.96
Total	29.36	53.99	83.35	

Year 11				
1	2.23	4.50	6.73	292.46
2	2.19	4.50	6.69	287.96
3	2.16	4.50	6.66	283.47
4	2.13	4.50	6.63	278.97
5	2.09	4.50	6.59	274.47
6	2.06	4.50	6.56	269.97
7	2.02	4.50	6.52	265.47
8	1.99	4.50	6.49	260.97
9	1.96	4.50	6.46	256.47
10	1.92	4.50	6.42	251.97
11	1.89	4.50	6.39	247.47
12	1.86	4.50	6.36	242.97
Total	24.50	53.99	78.4928	
Year 12				
1	1.82228	4.50	6.32	238.47
2	1.78853	4.50	6.29	233.97
3	1.75478	4.50	6.25	229.47
4	1.72104	4.50	6.22	224.97
5	1.68729	4.50	6.19	220.47
6	1.65355	4.50	6.15	215.97
7	1.6198	4.50	6.12	211.47
8	1.58605	4.50	6.09	206.97
9	1.55231	4.50	6.05	202.48
10	1.51856	4.50	6.02	197.98
11	1.48482	4.50	5.98	193.48
12	1.45107	4.50	5.95	188.98
Total	19.6401	53.99	73.6334	
Year 13				
1	1.45107	4.50	5.95	184.48
2	1.38358	4.50	5.88	179.98
3	1.34983	4.50	5.85	175.48
4	1.31609	4.50	5.82	170.98
5	1.28234	4.50	5.78	166.48
6	1.2486	4.50	5.75	161.98
7	1.21485	4.50	5.71	157.48
8	1.1811	4.50	5.68	152.98
9	1.14736	4.50	5.65	148.48
10	1.11361	4.50	5.61	143.98
11	1.07987	4.50	5.58	139.48
12	1.04612	4.50	5.55	134.98
Total	14.8144	53.99	68.8078	
Year 14				
1	1.01238	4.50	5.51	130.48
2	0.97863	4.50	5.48	125.98
3	0.94488	4.50	5.44	121.49
4	0.91114	4.50	5.41	116.99
5	0.87739	4.50	5.38	112.49
6	0.84365	4.50	5.34	107.99
7	0.8099	4.50	5.31	103.49

8	0.77615	4.50	5.28	98.99
9	0.74241	4.50	5.24	94.49
10	0.70866	4.50	5.21	89.99
11	0.67492	4.50	5.17	85.49
12	0.64117	4.50	5.14	80.99
Total	9.92128	53.99	63.9146	
Year 15				
1	0.60743	4.50	5.11	76.49
2	0.57368	4.50	5.07	71.99
3	0.53993	4.50	5.04	67.49
4	0.50619	4.50	5.01	62.99
5	0.47244	4.50	4.97	58.49
6	0.4387	4.50	4.94	53.99
7	0.40495	4.50	4.90	49.49
8	0.3712	4.50	4.87	44.99
9	0.33746	4.50	4.84	40.50
10	0.30371	4.50	4.80	36.00
11	0.26997	4.50	4.77	31.50
12	0.23622	4.50	4.74	27.00
Total	5.06188	53.99	59.0552	

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17	Year 18	Year 19	Year 20	Year 21	Year 22	Year 23	Year 24	Year 25
	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36	2036-37	2037-38	2038-39	2039-40	2040-41	2041-42	2042-43	2043-44	2044-45	2045-46	2046-47
Sales	588.46	609.06	630.37	652.44	675.27	698.91	723.37	748.69	774.89	802.01	830.08	859.14	889.20	920.33	952.54	985.88	1020.38	1056.10	1093.06	1131.32	1170.91	1211.90	1254.31	1298.21	1343.65
Total	588.46	609.06	630.37	652.44	675.27	698.91	723.37	748.69	774.89	802.01	830.08	859.14	889.20	920.33	952.54	985.88	1020.38	1056.10	1093.06	1131.32	1170.91	1211.90	1254.31	1298.21	1343.65
Expenditure																									
Raw Material Consumed	302.94	318.09	333.99	350.69	368.23	386.64	405.97	426.27	447.58	469.96	493.46	518.13	544.04	571.24	599.80	629.79	661.28	694.34	729.06	765.51	803.79	843.98	886.18	930.49	977.01
Water requirement	6.00	6.30	6.62	6.95	7.29	7.66	8.04	8.44	8.86	9.31	9.77	10.26	10.78	11.31	11.88	12.47	13.10	13.75	14.44	15.16	15.92	16.72	17.55	18.43	19.35
O & M Exps.	65.00	67.50	70.09	72.78	75.57	78.48	81.49	84.62	87.87	91.24	94.75	98.38	102.16	106.09	110.16	114.39	118.78	123.34	128.08	133.00	138.10	143.41	148.91	154.63	160.57
Rent, Taxes, Insurance	6.53	6.23	5.93	5.66	5.39	5.14	4.90	4.67	4.45	4.25	4.05	3.86	3.68	3.51	3.34	3.19	3.12	3.06	3.00	2.94	2.88	2.82	2.77	2.71	2.66
Cost of Production (A)	380.47	398.11	416.63	436.07	456.49	477.91	500.40	524.00	548.77	574.76	602.02	630.64	660.65	692.14	725.18	759.84	796.28	834.50	874.58	916.61	960.69	1006.93	1055.41	1106.26	1159.59
Administrative Expenses	9.00	9.45	9.92	10.42	10.94	11.49	12.06	12.66	13.30	13.96	14.66	15.39	16.16	16.97	17.82	18.71	19.65	20.63	21.66	22.74	23.88	25.07	26.33	27.64	29.03
Preoperative Exps w.off	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34	1.34
Cost of Sales (B)	10.34	10.79	11.26	11.76	12.28	12.82	13.40	14.00	14.63	15.30	16.00	16.73	17.50	18.31	19.16	20.05	20.98	21.97	23.00	24.08	25.22	26.41	27.66	28.98	30.36
Financial Charges																									
Interest on Term Loan	72.38	68.23	63.37	58.52	53.66	48.80	43.94	39.08	34.22	29.36	24.50	19.64	14.81	9.92	5.06										
Interest on Working Capital	10.59	10.98	11.38	11.79	12.22	12.67	13.13	13.61	14.10	14.62	15.15	15.70	16.28	16.87	17.49	18.13	18.79	19.47	20.19	20.92	21.69	22.48	23.31	24.16	25.05
Profit Before Depreciation & Tax	114.67	120.95	127.73	134.30	140.63	146.71	152.50	158.00	163.17	167.98	172.41	176.43	179.96	183.08	185.65	187.86	184.33	180.16	175.30	169.70	163.31	156.07	147.93	138.81	128.65
Depreciation	60.99	58.14	55.43	52.84	50.37	48.02	45.78	43.64	41.60	39.66	37.81	36.04	34.36	32.75	31.22	29.75	28.49	27.24	26.00	24.76	23.52	22.29	21.07	19.84	18.63
Profit Before Tax	53.68	62.81	72.30	81.46	90.26	98.69	106.73	114.36	121.57	128.32	134.60	140.39	145.60	150.33	154.43	157.12	171.84	167.91	163.30	157.94	151.79	144.78	136.86	127.97	118.02
Income Tax	20.03	21.13	22.31	23.46	24.57	25.63	26.64	27.60	28.51	29.35	30.12	30.82	31.44	31.98	32.43	32.82	32.20	31.47	30.62	29.65	28.53	27.27	25.84	24.25	22.48
Profit Transferred to Capital A/c	33.65	41.68	49.99	58.00	65.69	73.06	80.09	86.76	93.06	98.98	104.48	109.56	114.16	118.35	121.99	142.30	139.64	136.44	132.68	128.30	123.26	117.52	111.02	103.72	95.55
Cost of Generation	6.36	6.49	6.63	6.79	6.95	7.13	7.33	7.54	7.76	8.01	8.26	8.54	8.84	9.15	9.48	9.63	10.08	10.55	11.05	11.57	12.11	12.68	13.28	13.91	14.56
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.23	0.20	0.18	0.16	0.15	0.13	0.12	0.10	0.09	0.08	0.07	0.07
	6.36	5.80	5.29	4.83	4.42	4.05	3.71	3.41	3.14	2.89	2.66	2.46	2.27	2.10	1.94	1.76	1.64	1.54	1.44	1.34	1.26	1.17	1.10	1.03	0.96
Levelized cost of Gen	7.52																					13.52	13.87	14.22	14.56
Return of Equity (ROE)	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	58.87	74.70	74.70	74.70	74.70	74.70

Table 17: Projected Profit and Loss Statement

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17	Year 18	Year 19	Year 20	Year 21	Year 22	Year 23	Year 24	Year 25
(Rs. in Lacs)	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36	2036-37	2037-38	2038-39	2039-40	2040-41	2041-42	2042-43	2043-44	2044-45	2045-46	2046-47
Profit after Tax	33.65	41.68	49.99	58.00	65.69	73.06	80.09	86.76	93.06	98.98	104.48	109.56	114.16	118.35	121.99	142.30	139.64	136.44	132.68	128.30	123.26	117.52	111.02	103.72	95.55
Add :- Depreciation	60.99	58.14	55.43	52.84	50.37	48.02	45.78	43.64	41.60	39.66	37.81	36.04	34.36	32.75	31.22	12.75	12.49	12.24	12.00	11.76	11.52	11.29	11.07	10.84	10.63
Add :- Interest on Term Loan	72.38	68.23	63.37	58.52	53.66	48.80	43.94	39.08	34.22	29.36	24.50	19.64	14.81	9.92	5.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Add :- Interest on Working Capital Loan	10.59	10.98	11.38	11.79	12.22	12.67	13.13	13.61	14.11	14.62	15.15	15.70	16.28	16.87	17.49	18.13	18.79	19.48	20.19	20.93	21.69	22.48	23.31	24.16	25.05
Sub Total (A)	177.62	179.03	180.17	181.15	181.94	182.54	182.93	183.08	182.98	182.61	181.94	180.95	179.61	177.89	175.77	173.17	170.92	168.16	164.86	160.98	156.47	151.29	145.39	138.72	131.22
Interest on Term Loan	72.38	68.23	63.37	58.52	53.66	48.80	43.94	39.08	34.22	29.36	24.50	19.64	14.81	9.92	5.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interest on Working Capital	10.59	10.98	11.38	11.79	12.22	12.67	13.13	13.61	14.10	14.62	15.15	15.70	16.28	16.87	17.49	18.13	18.79	19.47	20.19	20.92	21.69	22.48	23.31	24.16	25.05
Repaym. of Term Loan	27.00	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sub Total (B)	109.98	133.21	128.75	124.30	119.87	115.46	111.06	106.68	102.32	97.97	93.64	89.34	85.09	80.79	76.54	18.13	18.79	19.47	20.19	20.92	21.69	22.48	23.31	24.16	25.05
D.S.C.R	1.62	1.34	1.40	1.46	1.52	1.58	1.65	1.72	1.79	1.86	1.94	2.03	2.11	2.20	2.30	9.55	9.10	8.63	8.17	7.69	7.21	6.73	6.24	5.74	5.24
cash flow		177.62	179.03	180.17	181.15	181.94	182.54	182.93	183.08	182.98	182.61	181.94	180.95	179.61	177.89	175.77	173.17	170.92	168.16	164.86	160.98	156.47	151.29	145.39	138.72

Table 18: Debt Service Coverage and Debt Equity Ratio

Table 19: Schedule of Administrative Expenses

Particulars	Nos.	Exps./ Month	Total Wages (Rs. In Lacs)
Other administrative exps.			
Telephone exps.		15000	1.80
Office exps.		30000	3.60
Conveyance		30000	3.60
Total			9.00

SCHEDULE - O																									
Particulars	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17	Year 18	Year 19	Year 20	Year 21	Year 22	Year 23	Year 24	Year 25
(Rs. in Lacs)	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36	2023-37	2023-38	2023-39	2023-40	2023-41	2023-42	2023-43	2023-44	2023-45	2023-46	2023-47
Sales	588.46	609.06	630.37	652.44	675.27	698.91	723.37	748.69	774.89	802.01	830.08	859.14	889.20	920.33	952.54	985.88	1020.38	1056.10	1093.06	1131.32	1170.91	1211.90	1254.31	1298.21	1343.65
If Sales is reduced by	8.00%																								
Cash Available as per DSCR sch M (E)	177.62	179.03	180.17	181.15	181.94	182.54	182.93	183.08	182.98	182.61	181.94	180.95	179.61	177.89	175.77	173.17	170.92	168.16	164.86	160.98	156.47	151.29	145.39	138.72	131.22
Change in Sales Value (F)	47.08	48.72	50.43	52.19	54.02	55.91	57.87	59.89	61.99	64.16	66.41	68.73	71.14	73.63	76.20	78.87	81.63	84.49	87.44	90.51	93.67	96.95	100.34	103.86	107.49
Revised Cash Profit (G=E-F)	130.54	130.31	129.74	128.95	127.92	126.63	125.06	123.19	120.99	118.45	115.53	112.22	108.48	104.26	99.56	94.30	89.29	83.67	77.41	70.47	62.80	54.34	45.05	34.86	23.73
Interest on TL (H)	72.38	68.23	63.37	58.52	53.66	48.80	43.94	39.08	34.22	29.36	24.50	19.64	14.81	9.92	5.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
TL Repayment (I)	27.00	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Debt (J=H+I)	99.38	95.23	117.37	112.51	107.65	102.79	97.93	93.07	88.21	83.35	78.49	73.63	68.81	63.91	59.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Revised DSCR (E/J)	1.31	1.37	1.11	1.15	1.19	1.23	1.28	1.32	1.37	1.42	1.47														
Revised Avg. DSCR	1.29																								
Purchases	302.9	318.1	334.0	350.7	368.2	386.6	406.0	426.3	447.6	470.0	493.5	518.1	544.0	571.2	599.8	629.8	661.3	694.3	729.1	765.5					
If Material price is increase by	12.00%																								
Cash Available as per DSCR sch M (E)	177.62	179.03	180.17	181.15	181.94	182.54	182.93	183.08	182.98	182.61	181.94	180.95	179.61	177.89	175.77	173.17	170.92	168.16	164.86	160.98					
Change in Purchase Value (F)	36.35	38.17	40.08	42.08	44.19	46.40	48.72	51.15	53.71	56.40	59.21	62.18	65.28	68.55	71.98	75.57	79.35	83.32	87.49	91.86					
Revised Cash Profit (G=E-F)	141.27	140.86	140.09	139.06	137.76	136.15	134.21	131.93	129.27	126.22	122.73	118.77	114.33	109.34	103.79	97.60	91.56	84.84	77.37	69.12					
Interest on TL (H)	72.38	68.23	63.37	58.52	53.66	48.80	43.94	39.08	34.22	29.36	24.50	19.64	14.81	9.92	5.06	0.00	0.00	0.00	0.00	0.00					
TL Repayment (I)	27.00	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	53.99	27.00	53.99	53.99	53.99	53.99					
Total Debt (J=H+I)	99.38	122.22	117.36	112.51	107.65	102.79	97.93	93.07	88.21	83.35	78.49	73.63	68.80	63.91	59.05	0.00	0.00	0.00	0.00	0.00					
Revised DSCR (G/J)	1.42	1.15	1.19	1.24	1.28	1.32	1.37	1.42	1.47	1.51	1.56														
Revised Avg. DSCR	1.36																								
Revised We. Average	1.71																								

Table 20: Sensitivity Analysis

Particulars/Year s	1st Year	2nd Year	3rd Year	4th Year	5th Year	6th Year	7th Year	8th Year	9th Year	10th Year	11th Year	12th Year	13th Year	14th Year	15th Year	16th Year	17th Year	18th Year	19th Year	20th Year	21st Year	22nd Year	23rd Year	24th Year	25th Year
Operating Profit	114.67	120.95	127.73	134.30	140.63	146.71	152.50	158.00	163.17	167.98	172.41	176.43	179.96	183.08	185.65	187.86	184.33	180.16	175.30	169.70	163.31	156.07	147.93	138.81	128.65
Add:- Depreciation (SLM)	60.99	58.14	55.43	52.84	50.37	48.02	45.78	43.64	41.60	39.66	37.81	36.04	34.36	32.75	31.22	29.75	28.24	26.78	25.36	23.98	22.64	21.34	20.07	18.83	17.63
Sub-Total	175.66	179.09	183.16	187.14	191.00	194.73	198.28	201.64	204.77	207.64	210.22	212.47	214.32	215.83	216.87	200.61	196.82	192.40	187.30	181.46	174.83	167.37	159.00	149.66	139.28
Less:- Depreciation (WDV)	449.55	273.50	167.66	103.97	65.58	42.37	28.29	19.69	14.39	11.07	8.96	7.57	6.62	5.94	5.43	5.03	4.70	4.42	4.16	3.93	3.70	3.51	3.33	3.16	3.00
Gross Income	-273.89	-94.41	15.50	83.17	125.42	152.35	169.99	181.95	190.38	196.56	201.26	204.90	207.70	209.89	211.44	195.58	192.12	187.98	183.13	177.53	171.13	163.85	155.66	146.49	136.28
Less:- Brought Forward of Losses	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	201.26	204.90	207.70	209.89	211.44	195.58	192.12	187.98	183.13	177.53	171.13	163.85	155.66	146.49	136.28
Less:- Deduction u/s 80IA ie. 100%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Net Taxable Income	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	201.26	204.90	207.70	209.89	211.44	195.58	192.12	187.98	183.13	177.53	171.13	163.85	155.66	146.49	136.28
Provision For Taxation (34.94%) (A)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70.32	71.59	72.57	73.34	73.88	68.33	67.13	65.68	63.99	62.03	59.79	57.25	54.39	51.18	47.62
Taxation Under MAT @ 17.47% (B)	20.03	21.13	22.31	23.46	24.57	25.63	26.64	27.60	28.51	29.35	30.12	30.82	31.44	31.98	32.43	32.82	32.20	31.47	30.62	29.65	28.53	27.27	25.84	24.25	22.48
Provision For Tax (A) or (B) which ever is higher	20.03	21.13	22.31	23.46	24.57	25.63	26.64	27.60	28.51	29.35	70.32	71.59	72.57	73.34	73.88	68.33	67.13	65.68	63.99	62.03	59.79	57.25	54.39	51.18	47.62
Credit of MAT U/S 115 JB	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	40.20	40.77	41.13	41.35	41.44	35.52	34.92	34.21	33.36	32.38	31.26	29.98	28.54	26.93	25.14
Provision for tax for the year	20.03	21.13	22.31	23.46	24.57	25.63	26.64	27.60	28.51	29.35	30.12	30.82	31.44	31.98	32.43	32.82	32.20	31.47	30.62	29.65	28.53	27.27	25.84	24.25	22.48

Table 21: Taxation Schedule

Table 22: Depreciation Schedule as per WDV Model

As Per WDV Model IT act				
Particulars	Land, Buildings & Civil Work	Plant & Machinery	Other Fixed Assets	Total
Rate of Depreciation	5%	40%	10%	
Opening Balance	200.00	1090.00	8.00	1298.00
Add :- Interest during construction period	1.23	6.72	0.05	8.00
Total	201.23	1096.72	8.05	1306.00
Year 1				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	10.06	438.69	0.81	449.55
W.D.V.	191.17	658.03	7.25	856.45
Year 2				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	9.56	263.21	0.72	273.50
W.D.V.	181.61	394.82	6.52	582.95
Year 3				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	9.08	157.93	0.65	167.66
W.D.V.	172.53	236.89	5.87	415.29
Year 4				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	8.63	94.76	0.59	103.97
W.D.V.	163.90	142.13	5.28	311.32
Year 5				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	8.20	56.85	0.53	65.58
W.D.V.	155.71	85.28	4.75	245.74
Year 6				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	7.79	34.11	0.48	42.37
W.D.V.	147.92	51.17	4.28	203.37
Year 7				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	7.40	20.47	0.43	28.29
W.D.V.	140.53	30.70	3.85	175.08
Year 8				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	7.03	12.28	0.39	19.69
W.D.V.	133.50	18.42	3.47	155.39

Year 9				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	6.68	7.37	0.35	14.39
W.D.V.	126.83	11.05	3.12	141.00
Year 10				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	6.34	4.42	0.31	11.07
W.D.V.	120.48	6.63	2.81	129.92
Year 11				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	6.02	2.65	0.28	8.96
W.D.V.	114.46	3.98	2.53	120.96
Year 12				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	5.72	1.59	0.25	7.57
W.D.V.	108.74	2.39	2.27	113.40
Year 13				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	5.44	0.95	0.23	6.62
W.D.V.	103.30	1.43	2.05	106.78
Year 14				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	5.16	0.57	0.20	5.94
W.D.V.	98.13	0.86	1.84	100.84
Year 15				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	4.91	0.34	0.18	5.43
W.D.V.	93.23	0.52	1.66	95.40
Year 16				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	4.66	0.21	0.17	5.03
W.D.V.	88.57	0.31	1.49	90.37
Year 17				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	4.43	0.12	0.15	4.70
W.D.V.	84.14	0.19	1.34	85.67
Year 18				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	4.21	0.07	0.13	4.42
W.D.V.	79.93	0.11	1.21	81.25
Year 19				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	4.00	0.04	0.12	4.16

W.D.V.	75.93	0.07	1.09	77.09
Year 20				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	3.80	0.03	0.11	3.93
W.D.V.	72.14	0.04	0.98	73.16
Year 21				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	3.61	0.00	0.10	3.70
W.D.V.	68.53	0.00	0.88	69.41
Year 22				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	3.43	0.00	0.09	3.51
W.D.V.	65.10	0.00	0.79	65.90
Year 23				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	3.26	0.00	0.08	3.33
W.D.V.	61.85	0.00	0.71	62.56
Year 24				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	3.09	0.00	0.07	3.16
W.D.V.	58.76	0.00	0.64	59.40
Year 25				
Additions during the year	0.00	0.00	0.00	0.00
Depreciation	2.94	0.00	0.06	3.00
W.D.V.	55.82	0.00	0.58	56.40

9.4.1 Cash Flow and profitability Analysis

It is planned to install of 1.2 Mwe Bamboo Based Gasifier Systems along with 4 Producer Gas Engines. The capacity utilisation in first year was assumed as 85%, and 85% from second year onwards. The cost of conversion of bamboo waste to pellet is not considered. This Capacity utilisation is based on 330 working days in a year.

Financial analysis is carried out with the following assumptions:

1. Bamboo Pellets are available at Rs.3 per kg. The cost of bamboo waste pellets increases by 5% every year.
2. Selling price of power is taken as Rs. 7.77 per unit. Increase in selling price is 3.5% per annum.
3. Operating hours are 8766 per annum
4. Depreciation @4.67% per annum
5. Interest @9% on the amount of loan from financial institutions.
6. Bamboo Consumption is 1.20 kg per unit.
7. 10% of power generated is consumed as auxiliary power and 2% is line loss.
8. Capital subsidy of Rs. 180.00 lakhs
9. Loan Repayment period of 15years.
10. Consumables are taken as Rs. 380.47 lakh in the first year.
11. Selling expenses are taken as Rs. 588.46 Lakh per year.
12. O & M charges are increasing by 3.84% every year.

Computation of Central Financial Assistance (CFA)

CFA is calculated as per the latest administrative approval of Ministry of New and Renewable Energy, Govt. of India. As per MNRE guidelines, CFA is Rs. 15.000 lakhs per 100 kW power generation using producer gas engines. Hence Capital subsidy is Rs. 180 lakhs for 1200 kW project.

RISK AND SENSITIVITY ANALYSIS

10.1 SENSITIVITY ANALYSIS

Sensitivity analysis is carried out with three cases namely:

- (1) Raw material cost increased by 10%.
- (2) Decrease in revenue by 10%.
- (3) Increase in raw material cost by 10% and decrease in revenue by 10%.

10.2 SAFETY ASPECTS

Statutory regulations on safety measures will be strictly followed. Safety appliances, viz extinguishers, sand buckets, earth rods, gloves, rubber mats, danger sign boards, safety regulation charts, etc. will be procured and installed as per safety norms. The factory will be insured for hazards. Protection equipment will be installed as per the Technical Advisory Committee of Insurance companies.

10.3 SWOT ANALYSIS

10.3.1 Strengths (S)

- a) Good engineering and technical skills to plan and build the project
- b) Bamboo residue waste is generated from bamboo processing unit. There will not be any shortage of raw material.
- c) Excellent support from the state government
- d) Availability of trained manpower and basic infrastructure
- e) Possibility of availing Carbon credits through CDM

10.3.2 Weaknesses (W)

- a) No experience of operating MW sized plant
- b) Limited experience of biomass gasification power generation through 100% producer gas engines.

10.3.3 Opportunities (O)

- a) As per the latest Electricity Act RE based grid connected power plant are encouraged to meet RE targets. This system will give quality and uninterrupted power at lower cost and address waste utilisation.
- b) Saves foreign exchange.
- c) Reduces the transmission losses of the Distribution Company.

10.3.4 Threats (T)

- a) Fluctuations in Bamboo waste costs.
- b) Variations in Biomass feed in tariff will affect the viability.

ANNEXURE 1

Fig. 7 HEAT & MASS BALANCE DIAGRAM

ANNEXURE 2

Fig. 8 OVERALL PLANT LAYOUT

ANNEXURE 3

Fig.9 P&I DIAGRAMS

ANNEXURE 4

SCOPE OF SUPPLY

i. Gas genset (300 KW)

Make	CNPC JICHAI POWER COMPANY LIMITED
Rated	300 Kw
Quantity	4 Nos
Phase	3 phase
Voltage	420

ii. Raw Material Handling

a. Hopper with Structure (100 HP1 & 100HP2) : For storage of biomass and feeding to belt conveyor

Hopper Capacity	1 Tonne
Nos. of Hoppers	2 Nos.
MOC	M.S. IS: 2062 (ST42), E-250

b. Belt conveyor (100 BC1 & 100BC2) : The belt conveyor from hopper to bucket elevator

Size inclined	800 mm x 6 metres
Power	2 HP
Quantity	2 Nos.

c. Bucket elevator (100 BE1& 100 BE2) : Carrying the biomass from belt conveyor to gasifier feed point.

Capacity	1000 Kg/hr
Power	3 HP
Quantity	2 Nos.
MOC	M.S. IS: 2062 (ST42), E-250

d. Wire Mesh Conveyor (100 WMC1) : For carrying Ash to Hand Trolleys

e. Pay loader (Client Scope) : Payloader will be used to put raw material into the hoppers for feeding belt conveyors.

8.3.1 Scope of supply –Gasifier 2 x 750L/HE

a. Gasifier (100 RCT1 & 100 RCT2)

1. Feeding and Spreading system: Feed bunker for biomass feeding to gasifier MOC
M.S. IS: 2062 (ST42), E-250

2. Conical hopper with vibrator: Conical hopper is above the reactor for intermediate storage cum preheating.

MOC M.S. IS:2062 (ST42), E-250

3. Reactor: All gasification reactions take place in the reactor.

Shell insulated with refractory lining white heat K of 50 mm thickness.

MOC AISI SS-304/SS-310

4. Ash drum: Ash drum is for collection of ash from gasification and further disposal to ash ponds.

MOC AISI SS-304.

5. Grate: Grate ensures uniform discharge of ash and char from the reactor.

The grate is rotated with the help of gearbox and motor and runs at intermittent time intervals usually for 10 sec every 10 minutes. The control is by ON/ OFF.

Grate motor 3 HP

Grate speed 0.5 rpm

MOC AISI SS-310

6. Primary cyclone with dust collection Box (100 CY1 & 100 CY2)

This is for primary dust separation from gas by cyclonic action.

Qty. 2 Nos

MOC AISI SS-304

7. Primary Scrubber (100VS1 & 100VS3)

It is a venturi scrubber meant for cooling and cleaning of gas. The scrubber is fitted with a water nozzle for spraying in the gas flow. The gas on scrubbing is cleaned and cools down to around 70 deg Celcius before entering the secondary Venturi. (These temperatures can vary according to ambient temp.)

Pump Capacity with motor 7.5 HP

Qty. 2 Nos.

MOC M.S. IS: 2062 (ST42), E – 250 / AISI SS-304

8. Secondary Scrubber with cooling Tower (100VS2 & 100VS4)

It is a venturi scrubber meant for cooling of gas. The scrubber is fitted with a water nozzle for spraying in the gas flow. The gas on scrubbing is cleaned and cools down to around 35-40 deg C before entering the blower.

Pump Capacity with motor 7.5 HP

MOC	M.S. IS: 2062 (ST42), E-250
Qty.	2 Nos

OR**ESP (Electrostatic Precipitator)**

ESP is required for ultra-good clean quality of gas.

Power Max.	7.5 HP
MOC	M.S. IS: 2062 (ST42), E-250
Electrode	SS-316
Qty.	1 Nos

9. Secondary cyclone (100CY3 & 100CY4): This is for removing moisture from scrubbed gas by cyclonic action.

Qty.	2 Nos
MOC	M.S. IS: 2062 (ST42), E-250

10. Blowers with VFD driven (100 BL1 & 100 BL2): The blower is for making a steady flow of gas with required pressure for the engine.

Pressure	1400 mmwc
Flow	2200 Nm ³ /hr
Motor	40 HP
MOC	SS-304 for fan impeller only
MOC	M.S. IS: 2062 (ST42), E-250 for body and casing
Quantity	2

11. Moisture Separator Box (100 MSB1 & 100 MSB2): It is for collecting the remaining moisture in gas.

Qty.	2 No.
MOC	M.S. IS: 2062 (ST42), E-250/SS-304

12. Heat Exchanger with cooling tower (100 HE1): Heat exchanger with cooling tower will be used to cool down the gas by the means of water cooling to the wet bulb temperature.

13. Coarse filter with standby line (100 CF1, 100 CF2, 100 CF3,)

There are 3 nos of coarse filters (one in each line including stand by line). The filtering medium is rice husk char. Our filters are user friendly, fed from top and old filter media is removed from bottom. One line is in operation at a time. After choking, lines will be changed for cleaning the standby filter.

MOC	M.S. IS: 2062 (ST42), E – 250
Qty	3 Nos

14. Fine filter with standby line (100 FF1, 100 FF2, 100 FF3,)

There are 3 nos. of fine filters (2 in each line including stand by line). The filtering medium is wood peel. Each filter is equipped with a screw down mechanism for adjusting pressure drop of around 30 mmwc. One line is in operation at a time. After choking, lines will be changed for cleaning the standby filter.

MOC	M.S. IS: 2062 (ST42), E-250
Qty	3 Nos

15. Fabric filter with standby line (100 FABF1, 100 FABF2, 100 FABF3,)

There are 1 nos of fabric filters (1 in each line including stand by line). The syngas is made to pass through a fabric filter before entering to the engine. The fabric is standard type cotton and is easily replaceable for cleaning. One line is in operation at a time. After choking, lines will be changed for cleaning the standby filter.

MOC	M.S. IS: 2062 (ST42), E-250
Qty	3 Nos

16. Test Flare (100 TF1)

For checking the quality of gas produced by the gasifier.

17 Ash and char handling system

The system comprises the following.

- a) Mud pump 3HP – 1no. for circulating water in ash drum
- b) Screen – 1no. for screening out char & ash from water
- c) Wire mesh conveyor (included in raw material handling)

18. Electrical and Automation system

It comprises a MCC & PLC panel, Grid Synchronising panel, H.T. Panel & Transformer with instruments to measure and transmit temperature, pressure etc. for control and supervision of the gasification system.

ANNEXURE 5



Analyzing for a Sustainable Future

NOIDA TESTING LABORATORIES

(A Government Approved Testing Laboratory)

An ISO 9001:2015 & 45001:2018 (OHSAS) Certified Laboratory

MoEF & CC (Ministry of Environment, Forest & Climate Change), UPPCB & HSPCB Recognized Laboratory

+91-9313611642, 8510081921, 7503031145, 8527870572, 7503031146, 9999794369

TEST CERTIFICATE

Test Report of	Report Code	Date of Issue
Bamboo Trunk	BS-041221-01	10/12/2021

Issued To: M/s Anagram Systems.

Address: B-57, Sector-2, Noida, Gautam Buddha Nagar, Uttar Pradesh-201307.

ANALYSIS DATA

Sample Received on	: 04/12/2021
Sample Drawn By	: Submitted By Customer
Sample Quantity	: Half of One Node to Another
Analysis Duration	: 04/12/2021 TO 10/12/2021
Sample Description	: Bamboo Stem

TEST RESULT			
S. No.	Name of Parameters	Test Method	Results
Proximate Analysis % by mass (On Received Basis)			
1.	Ash (% by Mass)	IS:1350(P-1)1984,Reaff-2007 Guideline	1.02
2.	Volatile Matter (% by Mass)	IS:1350(P-1)1984,Reaff-2007 Guideline	75.2
3.	Fixed Carbon (% by Mass)	IS:1350(P-1)1984,Reaff-2007 Guideline	12.81
4.	Total Moisture Content (% by Mass)	IS:1350(P-1)1984,Reaff-2007 Guideline	7.0
5.	Inherent Moisture ((% by Mass)	IS:1350(P-1)1984,Reaff-2007 Guideline	3.8
Ultimate Analysis % by mass (On Oven Dry Basis)			
6.	Total Carbon Content (% by Mass)	IS:1350(P-IV/Sec-1)/2011	22.5
7.	Total Hydrogen Content(% by Mass)	IS:1350(P-IV/Sec-1)/2011	3.5
8.	Total Nitrogen Content(% by Mass)	IS:1350(P-IV/Sec-1)/2011	0.95
9.	Total Sulphur Content(% by Mass)	IS:1350(P-III) 2011	1.35
10.	Mineral Matter (% by Mass)	IS:1350(P-I) 1984,Reaff-2007 Guideline	8.52
11.	Oxygen Content(% by Mass)	By Difference	18.6
12.	Calorific Value (K cal/kg)	Bomb Calorimeter	4350

Notes:

1. The results given above are related to the tested sample, as received & mentioned parameters. The customer asked for the above tests only.
2. Responsibility of the Laboratory is limited to the invoiced amount only.
3. This test report will not be generated again, either wholly or in part, without prior written permission of the laboratory.
4. This test report will not be used for any publicity/legal purpose.
5. The test samples will be disposed off after two weeks from the date of issue of test report, unless until specified by the customer.

Checked by

Authorized Signatory

Laboratory : GT-20, Sector-117, Noida Gautam Budh Nagar - 201301

Branch Office :

HARIDWAR | RUDRAPUR | CHANDIGARH | DEHRADUN | PUNE

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